# Numerical analysis of superconducting phases in the extended Hubbard model with non-local pairing

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**Abstract**

[To be continued. . . ]

- Appendix on $s$-wave SC ghost simulations on HM
- Appendix on triplet pairing in MFT self-consistent scheme
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# Contents



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**List of symbols and abbreviations**

| | |
|---|---|
| AF | Antiferromagnetic |
| BCS | Bardeen-Cooper-Schrieffer (theory) |
| BZ | Brillouin zone |
| $\beta$ | Inverse temperature |
| $\Delta^{(\gamma)}$ | $\gamma$-wave component of the gap function |
| DoF | Degree of freedom |
| EHM | Extended Hubbard model |
| EV(s) | Expectation value(s) |
| FS | Fermi surface |
| HF | Hartree-Fock |
| HFP(s) | Hartree-Fock parameter(s) |
| HM | Hubbard model |
| $L_\ell$ | Number of lattice sites in direction $\ell$ |
| LRT | Linear Response Theory |
| MBZ | Magnetic Brillouin zone |
| MFT | Mean-Field Theory |
| NBZ | Nematic Brillouin zone |
| NN(s) | Nearest neighbor(s) |
| RWC(s) | Relevant Wick's contraction(s) |
| RMP(s) | Renormalized model parameter(s) |
| SC | Superconductor |
| SDW | Spin-density wave |
| SSB | Spontaneous symmetry breaking |
| $T_\mathrm{c}$ | Critical temperature |

# Introduction

This thesis project is about my favorite ice cream flavor. [To be continued...]

# Chapter 1

# Mean-field theory discussion of the EHM

This chapter is devoted to develop a Mean Field Theory (MFT) framework of the Extended Hubbard model (EHM) of Eq. (??),

$$\hat{H} = \underbrace{-t\sum_{\langle ij\rangle}\sum_{\sigma}\hat{c}^{\dagger}_{i\sigma}\hat{c}_{j\sigma}}_{\hat{H}_t} \underbrace{+U\sum_{i}\hat{n}_{i\uparrow}\hat{n}_{i\downarrow}}_{\hat{H}_U} \underbrace{-V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'}\hat{n}_{i\sigma}\hat{n}_{j\sigma'}}_{\hat{H}_V}$$

Mean Field Theory (MFT) is a widely used and simple theoretical tool, often sufficient to describe the leading orders in phase transition phenomena of Many-Body Physics. Here MFT is employed generically, in order to later discuss the effects of the non-local term $\hat{H}_V$. The latter, following the path of Bardeen-Cooper-Schrieffer (BCS) theory in describing conventional $s$-wave superconductivity. As will be thoroughly described, the lattice spatial structure directly influences the topology of the gap function, giving rise to anisotropic pairing.

## 1.1 Extended Hubbard model symmetries and Wick's theorem

The general aim of this project is to study the phase diagram of the EHM by comparing ground-state energies of different phases. The phases we consider in next chapters for the EHM are the normal state (N), the antiferromagnetic ordering (AF), given by a non-uniform distribution of charge in each spin sector, and the anisotropic superconducting (SC) phase, described by a uniformly distributed charge allowing for Cooper pairing instabilities.

At the end of this section, in Tab. 1.2 we summarize all symmetries with the relative operations. Throughout the entire thesis we will refer to specific SSB for given groups to establish a specific phase.

### 1.1.1 Spatial and topological symmetries

Let us start from the trivial topology of the lattice we are working on. The EHM of Eq. (??) is evidently translational invariant on both the $x$ and $y$ directions. We refer to these groups as $\mathrm{T}^{(x)}_1$, $\mathrm{T}^{(x)}_1$. The "1" subscript indicates the one-site translations under which the model is symmetric.

Evidently, the following statement holds:

$$\mathrm{T}^{(\ell)}_n \supset \mathrm{T}^{(\ell)}_{kn} \qquad k \in \mathbb{N}$$

A model which is invariant for $n$-sites translations in the $\ell$ direction, also is for $kn$-sites translations. Then we can break translation symmetry in a given direction reducing it to a smaller group. For example, the commensurate AF phase we analyze halves translational invariance,

$$\bigotimes_{\ell=x,y}\mathrm{T}^{(\ell)}_1 \overset{\mathrm{SSB}}{\to} \bigotimes_{\ell=x,y}\mathrm{T}^{(\ell)}_2$$

since it deals with two-site periodic spin-density waves (SDWs); insted the SC phase preserves translational invariance.

Another natural symmetry of the EHM is the four-fold point group $\mathrm{C}_4$ of $\pi/2$ rotations, expressing the square lattice property being mapped onto itself under said rotations. A similar property as for the translations holds for point groups,

$$\mathrm{C}_n \supset \mathrm{C}_{n/k} \qquad k, n/k \in \mathbb{N}$$

which expresses the fact that, if a system is topologically invariant under $n$-fold rotations and $k \in \mathbb{N}$ exists such that $n/k \in \mathbb{R}$, then the smaller point group is automatically included. As for translations, we can break the point group reducing it to a smaller subgroup, say

$$\mathrm{C}_4 \quad \overset{\mathrm{SSB}}{\rightarrow} \quad \mathrm{C}_2$$

This is what happens in general with the effective arising of anisotropic corrections to the kinetic hamiltonian due to non-local interactions, as will be discussed in Sec. **??**. For our purposes, we will leave $\mathrm{C}_4$ unbroken both in the AF and SC phases.

Finally, the system also is inversion-symmetric: let $\mathrm{I}_2$ be the inversion group on the two-dimensional square lattice, containing trivially just the inversion operator and the identity. In general, rotations and mirroring are different operations from a group-theoretical point of view. However, when it comes to the square lattice, operatively it holds

$$\mathrm{I}_2 \equiv \mathrm{C}_2$$

To rotate by $\pi$ is the same as to mirror the lattice. The inversion symmetry is then already included in $\mathrm{C}_4$, and we leave it unspecified.

### 1.1.2 Spin and charge symmetries

When it comes to Fermi algebra and operators, a large symmetry of the EHM is a global U(2) symmetry [2] given by the transformation

$$\hat{c}_{i\sigma} \rightarrow \sum_{\sigma'} \mathcal{U}_{\sigma\sigma'} \hat{c}_{i\sigma'} \quad \text{with} \quad \mathcal{U}_{\sigma\sigma'} \in \mathrm{U}(2) \tag{1.1}$$

The proof is trivial and follows from the fact that $\mathcal{U}^\dagger \mathcal{U} = \mathcal{U}\mathcal{U}^\dagger = \mathbb{I}$ [add proof?]. Note that the the following decomposition holds,

$$\mathrm{U}(2) = \mathrm{SU}(2) \otimes \mathrm{U}^{\mathrm{c}}(1) \tag{1.2}$$

which expresses the separate charge conservation and spin rotational symmetries. In detail:

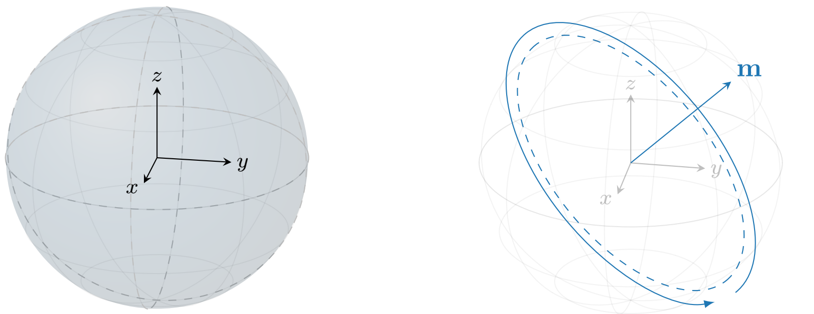
- $\mathrm{U}^{\mathrm{c}}(1)$ charge conservation, coming from Eq. (1.2) decomposition. By breaking this symmetry, we allow for the total charge to fluctuate in the ground state. These fluctuations are only allowed in the SC phase in the present text;

- SU(2) spin rotation symmetry, also , coming from Eq. (1.2) decomposition. Note that:

  $$\mathrm{SU}(2) \supset \mathrm{U}^{z}(1)$$

  The $z$ subscript we included to distinguish from the charge group. Hence SU(2) symmetry can be broken, selecting a particular vectorial direction for the order parameter (we call $z$), and reduced to a smaller $\mathrm{U}^{z}(1)$ symmetry which is essentially expressed by the conservation of the magnitude of the order parameter.

The last point is important: for the AF phase, for example, there is no need for the magnetization vector to be directed along the physical $z$-axis, which is the one perpendicular to the lattice plane. SU(2) symmetry breaks when (staggered) magnetization is established along a particular direction, but spin rotations around said direction still are ground state symmetries. This is shown schematically in Fig. 1.1.




a | SU(2) rotations domain before SSB.

b | U(1) rotations domain after SSB.

Figure 1.1 | Graphic representation of the concept of $\mathrm{SU}(2) \to \mathrm{U}^z(1)$ SSB. The solid color identifies, in both cases, the rotations domain for the current symmetry. When SSB takes place in a magnetic phase, the magnetization order parameter acquires a precise direction which was previously invariant over the full spherical domain. After SSB, the hamiltonian is rotationally invariant only for rotations *around* the selected magnetization vector.

| **Phase** | **SSBs** |
|---|---|
| AF | $\mathrm{T}_1^{(\ell)} \to \mathrm{T}_2^{(\ell)}$ |
| | $\mathrm{SU}(2) \to \mathrm{U}^z(1)$ |
| SC | $\mathrm{U}^c(1) \to 0$ |

Table 1.1 | SSB interpretation of the AF and SC phases. In the last line, the 0 on the SSB right-hand side indicates a completely disrupted symmetry.

### 1.1.3 Particle-hole symmetry at half-filling

The half-filled EHM is of particular interest due to its particle-hole symmetry (PHS), which is immediately distrupted at infinitesimal doping. Let us prove that. Consider the generic unitary operation

$$\mathrm{P} \quad : \quad \hat{c}_{i\sigma} \quad \to \quad e^{i\eta_{i\sigma}} \hat{h}^\dagger_{i\sigma}$$

being in general $\eta_i$ a phase dependent on the fermionic quantum state $(i\sigma)$. Inserting the above transformation in the EHM, we get

$$\mathrm{P}\{\hat{H}\} = -t \sum_{\langle ij \rangle} \sum_{\sigma} e^{i(\eta_{i\sigma}+\eta_{j\sigma})} \hat{h}_{i\sigma} \hat{h}^\dagger_{j\sigma} + U \sum_i \hat{h}_{i\uparrow} \hat{h}_{i\downarrow} \hat{h}^\dagger_{i\downarrow} \hat{h}^\dagger_{i\uparrow} - V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} \hat{h}_{i\sigma} \hat{h}_{j\sigma'} \hat{h}^\dagger_{j\sigma'} \hat{h}^\dagger_{i\sigma}$$

For the two interactions $\hat{H}_U$, $\hat{H}_V$ the phase factors cancelled out. We aim to find the conditions under which the above hamiltonian is mapped onto the EHM of (**??**),

$$\mathrm{P}\{\hat{H}\} \stackrel{?}{=} -t \sum_{\langle ij \rangle} \sum_{\sigma} \hat{h}^\dagger_{i\sigma} \hat{h}_{j\sigma} + U \sum_i \hat{h}^\dagger_{i\uparrow} \hat{h}^\dagger_{i\downarrow} \hat{h}_{i\downarrow} \hat{h}_{i\uparrow} - V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} \hat{h}^\dagger_{i\sigma} \hat{h}_{j\sigma'} \hat{h}_{j\sigma'} \hat{h}_{i\sigma}$$

Before starting the derivation, which is an extension of the argument of Arovas et al. [2] about the conventional HM, let us discuss some properties of the hole operators. Due to the presence of the phase factors in the PHS operative definition, the Fermi anticommutation rules hold regardlessly of the phase factor:

$$\begin{aligned} \{\hat{h}_{i\sigma}, \hat{h}^\dagger_{j\sigma'}\} &= e^{i(\eta_{i\sigma}-\eta_{j\sigma'})} \{\hat{c}^\dagger_{i\sigma}, \hat{c}_{j\sigma'}\} \\ &= e^{i(\eta_{i\sigma}-\eta_{j\sigma'})} \delta_{ij} \delta_{\sigma\sigma'} \\ &= \delta_{ij} \delta_{\sigma\sigma'} \end{aligned}$$

and evidently:

$$\hat{h}_{i\sigma} \hat{h}^\dagger_{i\sigma} = \hat{n}_{i\sigma}$$

being $\hat{n}_{i\sigma}$ the fermionic number operator on site $i$ with spin $\sigma$. It follows, for quartic interactions

$$\begin{aligned}
\hat{h}_{i\sigma}\hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'}\hat{h}^\dagger_{i\sigma} &= -\hat{h}_{i\sigma}\hat{h}_{j\sigma'}\hat{h}^\dagger_{i\sigma}\hat{h}^\dagger_{j\sigma'} \\
&= \hat{h}_{i\sigma}\hat{h}^\dagger_{i\sigma}\hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'} - \hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma'}\delta_{ij}\delta_{\sigma\sigma'} \\
&= -\hat{h}^\dagger_{i\sigma}\hat{h}_{i\sigma}\hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'} + \hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'} - \hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma'}\delta_{ij}\delta_{\sigma\sigma'} \\
&= \hat{h}^\dagger_{i\sigma}\hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma'}\hat{h}_{j\sigma'} - \hat{h}^\dagger_{i\sigma}\hat{h}_{i\sigma} + \hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'} - \hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma'}\delta_{ij}\delta_{\sigma\sigma'} \\
&= -\hat{h}^\dagger_{i\sigma}\hat{h}^\dagger_{j\sigma'}\hat{h}_{i\sigma}\hat{h}_{j\sigma'} - \hat{h}^\dagger_{i\sigma}\hat{h}_{i\sigma} + \hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'} - \left(\hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma'} - \hat{h}^\dagger_{i\sigma}\hat{h}_{j\sigma'}\right)\delta_{ij}\delta_{\sigma\sigma'} \\
&= \hat{h}^\dagger_{i\sigma}\hat{h}^\dagger_{j\sigma'}\hat{h}_{j\sigma'}\hat{h}_{i\sigma} + (\hat{n}_{i\sigma} + \hat{n}_{j\sigma'}) - 1 - \left(\hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma'} - \hat{h}^\dagger_{i\sigma}\hat{h}_{j\sigma'}\right)\delta_{ij}\delta_{\sigma\sigma'}
\end{aligned}$$

The first term of the last passage is mirrored with respect to the starting term. Now, let us deal with the kinetic, local and non-local interactions separately.

**Kinetic term.** Consider the kinetic term,

$$\mathrm{P}\{\hat{H}_t\} = -t \sum_{\langle ij \rangle} \sum_{\sigma} e^{i(\eta_{i\sigma} + \eta_{j\sigma})} \hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma}$$

It suffices to anticommute the two fermionic operators, and exchange site labels $i \leftrightarrow j$, to get

$$\mathrm{P}\{\hat{H}_t\} = -t \sum_{\langle ij \rangle} \sum_{\sigma} e^{i(\eta_{i\sigma} + \eta_{j\sigma} + \pi)} \hat{h}^\dagger_{i\sigma}\hat{h}_{j\sigma}$$

A necessary condition on the unitary operation P is the following,

$$\eta_{i\sigma} + \eta_{j\sigma} = (2k-1)\pi \quad \text{with} \quad k \in \mathbb{Z}$$

valid for NN sites.

**Local repulsion.** Substitute in $\hat{H}_U$ the above derivation,

$$\begin{aligned}
\mathrm{P}\{\hat{H}_U\} &= U \sum_i \hat{h}_{i\uparrow}\hat{h}_{i\downarrow}\hat{h}^\dagger_{i\downarrow}\hat{h}^\dagger_{i\uparrow} \\
&= U \sum_i \hat{h}^\dagger_{i\uparrow}\hat{h}^\dagger_{i\downarrow}\hat{h}_{i\downarrow}\hat{h}_{i\uparrow} + U \sum_i (\hat{n}_{i\uparrow} + \hat{n}_{i\downarrow}) - U \sum_i (1) \\
&= U \sum_i \hat{h}^\dagger_{i\uparrow}\hat{h}^\dagger_{i\downarrow}\hat{h}_{i\downarrow}\hat{h}_{i\uparrow} + U(\hat{N} - L_x L_y)
\end{aligned}$$

Thus, being for the half-filled state

$$\langle \hat{N} \rangle = L_x L_y$$

preserves PHS in the local sector.

**Non-local attraction.** Proceed identically for $\hat{H}_V$,

$$\begin{aligned}
\mathrm{P}\{\hat{H}_V\} &= -V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} \hat{h}_{i\sigma}\hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'}\hat{h}^\dagger_{i\sigma} \\
&= -V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} \hat{h}^\dagger_{i\sigma}\hat{h}^\dagger_{j\sigma'}\hat{h}_{j\sigma'}\hat{h}_{i\sigma} - V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} (\hat{n}_{i\sigma} + \hat{n}_{j\sigma'}) + V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} (1) \\
&= -V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} \hat{h}^\dagger_{i\sigma}\hat{h}^\dagger_{j\sigma'}\hat{h}_{j\sigma'}\hat{h}_{i\sigma} - 2zV(\hat{N} - L_x L_y)
\end{aligned}$$

with $z = 4$ the lattice coordination number. In the last passage, we used that on the square lattice

$$\begin{aligned}
\sum_{\langle ij \rangle} (1) &= \sum_{i \in \mathcal{S}/2} \sum_{j \in \mathrm{NN}(i)} (1) \\
&= \frac{L_x L_y}{2} \times z
\end{aligned} \tag{1.3}$$

| Symmetry | Operations |
|---|---|
| $\mathrm{T}_1^{(x)} \otimes \mathrm{T}_1^{(y)}$ | One-site discrete translations separately in both directions |
| $\mathrm{C}_4$ | Four-fold point group (real $\pi/2$ angle rotations) |
| $\mathrm{U}^{\mathrm{c}}(1)$ | Charge-conserving global phase shifts |
| $\mathrm{SU}(2)$ | Three-dimensional rotations in spin space |
| PHS | Particle-hole exchange (only at half filling) |
| $\mathrm{U}^{z}(1)$ | Reduced two-dimensional rotations in spin space |

Table 1.2 | Model symmetries and relative group operations. Note that $\mathrm{U}^{\mathrm{c}}(1)$ represents the charge conserving group, while $\mathrm{U}^{z}(1)$ represents the subgroup obtained by breaking the SU(2) symmetry with a vector order parameter and preserving its amplitude. Since the latter is "derived" from SU(2), we separate it from the four main symmetries of the model.

which will turn out quite often. As before, an half-filled phase preserves PHS in the non-local sector. In last passage we used the fact that summing over NNs means overcounting each site $z/2$ times, and spin degeneracy.

No other condition on phases is derived, thus we conclude the half-filled EHM preserves PHS if the two sublattices $\mathcal{S}_a$, $\mathcal{S}_b$ the square lattice can be divided into, being a (trivial) bipartite lattice, differ by a $\pi$ phase difference under PHS transform. In other words, if P is defined by the unitary operation

$$\mathrm{P} \quad : \quad \begin{cases} \hat{c}_{i\sigma} \to \hat{h}^\dagger_{i\sigma} & \text{for} \quad i \in \mathcal{S}_a \\ \hat{c}_{i\sigma} \to -\hat{h}^\dagger_{i\sigma} & \text{for} \quad i \in \mathcal{S}_b \end{cases}$$

and the system is half-filled, PHS is respected. Any other phase choice, as long as it alternates by $\pi$ on the two sublattices, is good enough. To summarize, we report in Tab. 1.2 all identified symmetries.

## 1.2 MFT on Hubbard lattices

This section serves as groundwork for everything that follows. We here define the MFT scheme as well as the iterative HF computational scheme we use to estimate parametrically the relevant physical quantities of the various phases. We start by recalling the general features of MFT and then move to the specificity of the EHM, apply Wick's theorem and finally move to algorithic aspects.

### 1.2.1 MFT general procedure, self-consistency and Wick's theorem

The Mean Field Theory approach to any fermionic quartic operator is given by the following procedure: given the hamiltonian made of the quadratic operators $\hat{A}$ and $\hat{B}$

$$\hat{H} \equiv \hat{A}\hat{B}$$

we define the fluctuations

$$\delta\hat{A} \equiv \hat{A} - \langle\hat{A}\rangle \qquad \delta\hat{B} \equiv \hat{B} - \langle\hat{B}\rangle$$

thus the following chain holds:

$$\begin{aligned} \hat{A}\hat{B} &= \langle\hat{A}\rangle\langle\hat{B}\rangle + \delta\hat{A}\langle\hat{B}\rangle + \langle\hat{A}\rangle\delta\hat{B} + \delta\hat{A}\delta\hat{B} \\ &= -\langle\hat{A}\rangle\langle\hat{B}\rangle + \hat{A}\langle\hat{B}\rangle + \langle\hat{A}\rangle\hat{B} + \underbrace{\delta\hat{A}\delta\hat{B}}_{\mathcal{O}(\delta)^2} \end{aligned}$$

which implies

$$\langle\hat{H}\rangle = \langle\hat{A}\rangle\langle\hat{B}\rangle + \langle\delta\hat{A}\delta\hat{B}\rangle$$

Thus, if $|\langle\hat{A}\rangle\langle\hat{B}\rangle| \gg |\langle\delta\hat{A}\delta\hat{B}\rangle|$, in a statistical sense we can neglect the $\mathcal{O}(\delta)^2$ part of the hamiltonian and approximate

$$\hat{H} \simeq \hat{A}\langle\hat{B}\rangle + \langle\hat{A}\rangle\hat{B} - \langle\hat{A}\rangle\langle\hat{B}\rangle \qquad \Longrightarrow \qquad \langle\hat{H}\rangle \simeq \langle\hat{A}\rangle\langle\hat{B}\rangle \tag{1.4}$$

| Term | Normal phase | Antiferromagnet | Superconductor |
|---|---|---|---|
| Hartree | | | |
| Fock | Prohibited | Prohibited | Prohibited |
| Cooper | Prohibited | Prohibited | |

Table 1.3 Summary of Wick's terms selection rules and relevant contributions to the effective hamiltonian for each phase limitedly to $\hat{H}_U$.

Self consistency of MFT procedure is thus embodied by the condition

$$\left|\frac{\langle \delta\hat{A}\delta\hat{B}\rangle}{\langle\hat{A}\rangle\langle\hat{B}\rangle}\right| \ll 1 \qquad \Longrightarrow \qquad \left|\frac{\langle\hat{A}\hat{B}\rangle}{\langle\hat{A}\rangle\langle\hat{B}\rangle} - 1\right| \ll 1 \tag{1.5}$$

which essentially states that the least the fluctuations, the better the approximation. This rule of thumb is necessary in order to interpret the physical relevance of numerical MFT results. We will come back to this in the end, for now assuming the validity of MFT approximation and deriving the mathematical results.

### 1.2.2 MFT treatment of the pure Hubbard model

The pure Hubbard model, $\hat{H}_t + \hat{H}_U$, offers a simple framework for MFT analysis of the AF phase: this is described in detail in App. A.1. The key passage is there given by the MFT approximation

$$\hat{n}_{i\uparrow}\hat{n}_{i\downarrow} \simeq \hat{n}_{i\uparrow}\langle\hat{n}_{i\downarrow}\rangle + \langle\hat{n}_{i\uparrow}\rangle\hat{n}_{i\downarrow} + (\text{constants}) \tag{1.6}$$

from which the AF structure is simply recovered. In terms of the discussion of the above paragraph, we have $\hat{A} \equiv \hat{n}_{i\uparrow}$ and $\hat{B} \equiv \hat{n}_{i\downarrow}$. However the "full" operator is given by

$$\hat{n}_{i\uparrow}\hat{n}_{i\downarrow} = \hat{c}^\dagger_{i\uparrow}\hat{c}^\dagger_{j\downarrow}\hat{c}_{j\downarrow}\hat{c}_{i\uparrow}$$

whose terms we can contract in three different ways following Wick's Theorem:

$$\langle\hat{c}^\dagger_{i\uparrow}\hat{c}^\dagger_{i\downarrow}\hat{c}_{i\downarrow}\hat{c}_{i\uparrow}\rangle \simeq \underbrace{\langle\hat{c}^\dagger_{i\uparrow}\hat{c}^\dagger_{i\downarrow}\rangle\langle\hat{c}_{i\downarrow}\hat{c}_{i\uparrow}\rangle}_{\text{Cooper}} - \underbrace{\langle\hat{c}^\dagger_{i\uparrow}\hat{c}_{i\downarrow}\rangle\langle\hat{c}^\dagger_{i\downarrow}\hat{c}_{i\uparrow}\rangle}_{\text{Fock}} + \underbrace{\langle\hat{c}^\dagger_{i\uparrow}\hat{c}_{i\uparrow}\rangle\langle\hat{c}^\dagger_{i\downarrow}\hat{c}_{i\downarrow}\rangle}_{\text{Hartree}} \tag{1.7}$$

where the $-$ sign in the Fock term comes from fermionic anticommutation rules.

As a first approximation, the theorem is assumed to hold (which, in a BCS-like fashion, is equivalent to assuming for the ground-state to be a coherent state). The AF ground state reduces translational invariance and rotational symmetry, as explained in Tab. 1.1. Thus, of the three terms above, when considering the pure Hubbard model in AF phase:

- Cooper fluctuations are suppressed, because they break charge conservation;
- Similarly the Fock term is null as well because if $i = j$ and $\sigma' = \overline{\sigma}$ (as is for the local interaction, which contains the operator $\hat{n}_{i\uparrow}\hat{n}_{i\downarrow}$) the expectation values involved are describing a process breaking the survivor $\mathrm{U}^z(1)$ spin symmetry.

Thus, correctly, the Wick's decomposition of pure HM only involves Hartree-terms of Eq. (1.7), getting us back to (1.6). The distinction between the normal phase and the AF phase is given by whether the Hartree fluctuations break or not translational symmetry: AF is indeed a specific SDW ordering which weakens the latter. In general the three parts of the decomposition (Hartree, Fock, Cooper) need to be considered altogether: this is what is done in the next chapters. What said here extends naturally to the EHM: the relevant contributions for each phase given by the local repulsion $\hat{U}$ in the EHM are listed in Tab. 1.3.

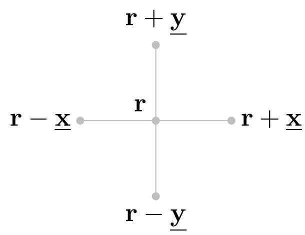



Figure 1.2 | Schematic representation of the four NNs of a given site $i$ for a planar square lattice.

## 1.3 The non-local attraction $\hat{H}_V$ as a source of symmetry-breaking interactions

Let us now extend the discussion to the EHM and consider the NN non-local attraction,

$$\hat{H}_V \equiv -V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma\sigma'} \hat{n}_{\mathbf{r}\sigma} \hat{n}_{\mathbf{r}'\sigma'} \tag{1.8}$$

where we changed notation, and replaced the site index $i$ with its 2D position on the lattice

$$i \quad \rightarrow \quad \mathbf{r} = (x, y)$$

Let also the lattice versors,

$$\underline{\mathbf{x}} \equiv (1, 0) \qquad \underline{\mathbf{y}} \equiv (0, 1)$$

as in Fig. 1.2.

Evidently the hamiltonian can be decomposed in various spin terms,

$$\begin{aligned} \hat{H}_V &= \sum_{\sigma\sigma'} \hat{H}_V^{\sigma\sigma'} \\ &= \underbrace{\hat{H}_V^{\uparrow\uparrow} + \hat{H}_V^{\downarrow\downarrow}}_{\text{Same-spin}} + \underbrace{\hat{H}_V^{\uparrow\downarrow} + \hat{H}_V^{\downarrow\uparrow}}_{\text{Opposite-spin}} \end{aligned} \tag{1.9}$$

The role of said terms is crucial in establishing the pairing channel of the dominant physical phase. As an example, the s.s. terms can be seen as a possible source of triplet pairing superconductivity. Next section are devoted to analyze said terms and derive a simple analytical result in reciprocal space.

### 1.3.1 Real space description

Let us start from the real space form of Eq. (1.9). To carry out a summation over nearest neighbors $\langle \mathbf{r}\mathbf{r}' \rangle$ of a square lattice means precisely to sum over all links of the lattice. Then we can identify the generic opposite-spin (o.s.) term $\hat{H}_V^{\sigma\overline{\sigma}}$ as the one collecting the $\sigma$ operators of sublattice $\mathcal{S}_a$ and $\overline{\sigma}$ operators of sublattice $\mathcal{S}_b$. The o.s. non-local interactions can be written as a sum of terms over just one of the two sublattices $\mathcal{S}_a$ and $\mathcal{S}_b$. Define the site hamiltonian

$$\hat{h}_V^{(\mathbf{r})} \equiv -V \sum_{\boldsymbol{\delta}} \left( \hat{n}_{\mathbf{r}\uparrow} \hat{n}_{\mathbf{r}+\boldsymbol{\delta}\downarrow} + \hat{n}_{\mathbf{r}\uparrow} \hat{n}_{\mathbf{r}-\boldsymbol{\delta}\downarrow} \right) \quad \text{with} \quad \boldsymbol{\delta} \in \{\underline{\mathbf{x}}, \underline{\mathbf{y}}\} \tag{1.10}$$

Then, it follows,

$$\begin{aligned} \hat{H}_V^{(\text{o.s.})} &= \overbrace{\sum_{\mathbf{r}\in\mathcal{S}_a} \hat{h}_V^{(\mathbf{r})}}^{\hat{H}_V^{\uparrow\downarrow}} + \overbrace{\sum_{\mathbf{r}\in\mathcal{S}_b} \hat{h}_V^{(\mathbf{r})}}^{\hat{H}_V^{\downarrow\uparrow}} \\ &= \sum_{\mathbf{r}\in\mathcal{S}} \hat{h}_V^{(\mathbf{r})} \end{aligned} \tag{1.11}$$

| Sector | Term | Normal phase | Antiferromagnet | Superconductor |
|---|---|---|---|---|
| o.s. | Hartree | | | |
| | Fock | Prohibited | Prohibited | Prohibited |
| | Cooper | Prohibited | Prohibited | |
| s.s. | Hartree | | | |
| | Fock | | | |
| | Cooper | Prohibited | Prohibited | |

Table 1.4 Summary of Wick's terms selection rules and relevant contributions to the effective hamiltonian for each phase limitedly to $\hat{H}_V$, assuming preservation of $\mathrm{U}^z(1)$ symmetry.

Here the notation of Fig. **??** is used. For each site $\mathbf{r}$ in a given sublattice, the nearest neighbors sites are four – all in the other sublattice. Similarly, the same-spin (s.s.) hamiltonian decomposes as

$$\hat{H}_V^{(\text{s.s.})} = -V \sum_{\mathbf{r}\in\mathcal{S}_a} \sum_{\boldsymbol{\delta},\sigma} \left(\hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\sigma} + \hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}-\boldsymbol{\delta}\sigma}\right) \quad \text{with} \quad \boldsymbol{\delta} \in \{\underline{\mathbf{x}}, \underline{\mathbf{y}}\}$$

**Opposite-spin terms.** Consider first the o.s. terms of Eq. (1.10): take e.g. the NNs term $\hat{n}_{i\uparrow}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\downarrow}$. As in Eq. (1.7), taking Wick's contractions,

$$\begin{aligned}\langle \hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\downarrow}\rangle &= \langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}+\boldsymbol{\delta}\downarrow}\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\rangle \\ &\simeq \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}+\boldsymbol{\delta}\downarrow}\rangle\langle \hat{c}_{\mathbf{r}+\boldsymbol{\delta}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\rangle}_{\text{Cooper}} - \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\downarrow}\rangle\langle \hat{c}^\dagger_{\mathbf{r}+\boldsymbol{\delta}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\uparrow}\rangle\langle \hat{c}^\dagger_{\mathbf{r}+\boldsymbol{\delta}\downarrow}\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\downarrow}\rangle}_{\text{Hartree}}\end{aligned}$$

Identical decompositions are given for all others NNs. Of the three terms above:

- The Cooper term breaks $\mathrm{U}^c(1)$ charge symmetry, allowing for supeconducting instabilities;
- The Fock term breaks $\mathrm{U}^z(1)$ symmetry, because it accounts for a site hop *plus* spin flip process;
- The Hartree term can reduce both translational invariance and spin rotational symmetry.

As a general rule of thumb, throughout this text we will discard the o.s. Fock term eveywhere, preserving a robust $\mathrm{U}^z(1)$ symmetry. For the normal state as well as for the AF instability only the Hartree term is to be accounted; for the SC phase also the Cooper term contributes. Note that for superconducting instabilities, due to superexchange mechanism (as explained in App. A) the o.s. term accounts for singlet pairing as well as zero-projection triplet pairing. Which channel is preferred, is a matter of thermodynamic advantage.

**Same-spin terms.** Consider then the same-spin terms: take e.g. the term $\hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\uparrow}$. As above,

$$\begin{aligned}\langle \hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\uparrow}\rangle &= \langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}+\boldsymbol{\delta}\uparrow}\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\uparrow}\hat{c}_{\mathbf{r}\uparrow}\rangle \\ &= \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}+\boldsymbol{\delta}\uparrow}\rangle\langle \hat{c}_{\mathbf{r}+\boldsymbol{\delta}\uparrow}\hat{c}_{\mathbf{r}\uparrow}\rangle}_{\text{Cooper}} - \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\uparrow}\rangle\langle \hat{c}^\dagger_{\mathbf{r}+\boldsymbol{\delta}\uparrow}\hat{c}_{\mathbf{r}\uparrow}\rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\uparrow}\rangle\langle \hat{c}^\dagger_{\mathbf{r}+\boldsymbol{\delta}\uparrow}\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\uparrow}\rangle}_{\text{Hartree}}\end{aligned}$$

Identical consideration as in the above paragraph hold for each term. The only difference with the o.s. terms is given by the Fock term: since the spin-flip process is absent, now the Fock fluctuations actually contribute as an effective NN hopping term. In other words, this term does not break $\mathrm{U}^z(1)$ symmetry and thus is perfectly legitimate in each phase. Tab. 1.4 summarizes the selection rules hereby presented.

### 1.3.2 Fourier transform of the model

It is useful to derive analytically the reciprocal-space form of the three hamiltonian parts. By the sign $\stackrel{\mathcal{F}}{=}$, we indicate a passage where a Fourier-transform is performed. $\mathcal{F}$-transform is obtained according to the convention Fourier transform it according to the convention

$$\hat{c}_{\mathbf{r}\sigma} \stackrel{\mathcal{F}}{=} \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k}\in\mathrm{BZ}} e^{-i\mathbf{k}\cdot\mathbf{r}} \hat{c}_{\mathbf{k}\sigma} \tag{1.12}$$

**Kinetic part.** The kinetic part is the easiest to transform. Apply the definition,

$$\begin{aligned}
-t \sum_{\langle\mathbf{r}\mathbf{r}'\rangle} \sum_{\sigma} \left[\hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}'\sigma} + \text{h.c.}\right] &\stackrel{\mathcal{F}}{=} -t \sum_{\langle\mathbf{r}\mathbf{r}'\rangle} \sum_{\sigma} \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k}\in\mathrm{BZ}} e^{i\mathbf{k}\cdot\mathbf{r}} \hat{c}^\dagger_{\mathbf{k}\sigma} \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k}'\in\mathrm{BZ}} e^{-i\mathbf{k}'\cdot\mathbf{r}'} \hat{c}_{\mathbf{k}'\sigma} + \text{h.c.} \\
&= -2t \sum_{\mathbf{k},\mathbf{k}'} \sum_{\sigma} \hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}'\sigma} \frac{1}{L_x L_y} \sum_{\mathbf{r}\in\mathcal{S}/2} e^{i(\mathbf{k}-\mathbf{k}')\cdot\mathbf{r}} \left[e^{ik'_x} + e^{-ik'_x} + e^{ik'_y} + e^{-ik'_y}\right] \\
&= \sum_{\mathbf{k}\in\mathrm{BZ}} \sum_{\sigma} \epsilon_{\mathbf{k}} \hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma}
\end{aligned} \tag{1.13}$$

where the bare bands were defined,

$$\epsilon_{\mathbf{k}} \equiv -2t\left(\cos k_x + \cos k_y\right) \tag{1.14}$$

**Non-local attraction.** Consider a generic bond, say, the one connecting NNs $\mathbf{r}$ and $\mathbf{r}\pm\boldsymbol{\delta}$. Then:

$$\begin{aligned}
\hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}\pm\boldsymbol{\delta}\sigma'} &= \hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\sigma'}\hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\sigma'}\hat{c}_{\mathbf{r}\sigma} \\
&\stackrel{\mathcal{F}}{=} \frac{1}{(L_x L_y)^2} \sum_{\mathbf{k}_1,\cdots,\mathbf{k}_4} e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{r}} e^{\pm i(\mathbf{k}_2-\mathbf{k}_3)\cdot\boldsymbol{\delta}} \hat{c}^\dagger_{\mathbf{k}_1\sigma}\hat{c}^\dagger_{\mathbf{k}_2\sigma'}\hat{c}_{\mathbf{k}_3\sigma'}\hat{c}_{\mathbf{k}_4\sigma}
\end{aligned}$$

Then, the interaction at site $\mathbf{r}$, spin $\sigma$ with its NNs at spin $\sigma'$ – indicated as $(\mathbf{r}\sigma\sigma')$ – is given by

$$\begin{aligned}
(\mathbf{r}\sigma\sigma') &= -V \sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} \left[\hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\sigma'} + \hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}-\boldsymbol{\delta}\sigma'}\right] \\
&\stackrel{\mathcal{F}}{=} -\frac{V}{(L_x L_y)^2} \sum_{\boldsymbol{\delta}} \sum_{\mathbf{k}_1,\cdots,\mathbf{k}_4} e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{r}} \\
&\qquad\qquad \times \left(e^{i(\mathbf{k}_2-\mathbf{k}_3)\cdot\boldsymbol{\delta}} + e^{-i(\mathbf{k}_2-\mathbf{k}_3)\cdot\boldsymbol{\delta}}\right) \hat{c}^\dagger_{\mathbf{k}_1\sigma}\hat{c}^\dagger_{\mathbf{k}_2\sigma'}\hat{c}_{\mathbf{k}_3\sigma'}\hat{c}_{\mathbf{k}_4\sigma} \\
&= -\frac{2V}{(L_x L_y)^2} \sum_{\boldsymbol{\delta}} \sum_{\mathbf{k}_1,\cdots,\mathbf{k}_4} e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{r}} \cos\left[(\mathbf{k}_2-\mathbf{k}_3)\cdot\boldsymbol{\delta}\right] \hat{c}^\dagger_{\mathbf{k}_1\sigma}\hat{c}^\dagger_{\mathbf{k}_2\sigma'}\hat{c}_{\mathbf{k}_3\sigma'}\hat{c}_{\mathbf{k}_4\sigma}
\end{aligned}$$

The full non-local interaction is given by summing over all sites of $\mathcal{S}/2$ (which is, one sublattice of $\mathcal{S}$). This gives back momentum conservation,

$$\frac{1}{L_x L_y} \sum_{\mathbf{r}\in\mathcal{S}/2} e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{r}} = \frac{1}{2}\delta_{\mathbf{k}_1+\mathbf{k}_2=\mathbf{k}_3+\mathbf{k}_4}$$

Let $\mathbf{k}_1+\mathbf{k}_2 = \mathbf{k}_3+\mathbf{k}_4 = \mathbf{K}$, and define $\mathbf{k}$, $\mathbf{k}'$ such that

$$\mathbf{k}_1 \equiv \mathbf{K}+\mathbf{k} \qquad \mathbf{k}_2 \equiv \mathbf{K}-\mathbf{k} \qquad \mathbf{k}_3 \equiv \mathbf{K}-\mathbf{k}' \qquad \mathbf{k}_4 \equiv \mathbf{K}+\mathbf{k}' \qquad \delta\mathbf{k} \equiv \mathbf{k}-\mathbf{k}'$$

Sums over these variables must be intended as over the Brillouin Zone (BZ). Hence:

$$\begin{aligned}
\hat{H}_V &= \sum_{\mathbf{r}\in\mathcal{S}_a} \sum_{\sigma\sigma'} (\mathbf{r}\sigma\sigma') \\
&\stackrel{\mathcal{F}}{=} -\frac{V}{L_x L_y} \sum_{\sigma\sigma'} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \cos\left(\delta\mathbf{k}\cdot\boldsymbol{\delta}\right) \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma'}\hat{c}_{\mathbf{K}-\mathbf{k}'\sigma'}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} \\
&= -\frac{V}{L_x L_y} \sum_{\sigma\sigma'} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \left[\cos\left(\delta k_x\right) + \cos\left(\delta k_y\right)\right] \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma'}\hat{c}_{\mathbf{K}-\mathbf{k}'\sigma'}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}
\end{aligned} \tag{1.15}$$

which decomposes in s.s. and o.s. channels as:

$$\hat{H}_V \overset{\mathcal{F}}{=} \overbrace{-\frac{V}{L_xL_y}\sum_{\sigma}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\left[\cos\left(\delta k_x\right)+\cos\left(\delta k_y\right)\right]\hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}'\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}}^{\text{s.s.}}$$
$$\underbrace{-\frac{V}{L_xL_y}\sum_{\sigma}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\left[\cos\left(\delta k_x\right)+\cos\left(\delta k_y\right)\right]\hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\overline{\sigma}}\hat{c}_{\mathbf{K}-\mathbf{k}'\overline{\sigma}}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}}_{\text{o.s.}}$$

The o.s. part can be simplified: since

$$\hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\overline{\sigma}}\hat{c}_{\mathbf{K}-\mathbf{k}'\overline{\sigma}}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} = \hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\overline{\sigma}}\hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}'\overline{\sigma}}$$

due to fermionic anti-commutation rules, and since the composite transform

$$\mathbf{k} \to -\mathbf{k} \qquad \mathbf{k}' \to -\mathbf{k}'$$

leaves the form factor $\cos\left(\delta k_x\right)+\cos\left(\delta k_y\right)$ invariant, we reduce the spin sum to two times the $\sigma =\uparrow$ part,

$$\hat{H}_V^{(\text{o.s.})} = -\frac{2V}{L_xL_y}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\left[\cos\left(\delta k_x\right)+\cos\left(\delta k_y\right)\right]\hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\uparrow}\hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\downarrow}\hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow}\hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \tag{1.16}$$

whiel the explicit s.s. sector is given immediately by:

$$\hat{H}_V^{(\text{s.s.})} = -\frac{V}{L_xL_y}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\left[\cos\left(\delta k_x\right)+\cos\left(\delta k_y\right)\right]$$
$$\times\left[\hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\uparrow}\hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\uparrow}\hat{c}_{\mathbf{K}-\mathbf{k}'\uparrow}\hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow}+\hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\downarrow}\hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\downarrow}\hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow}\hat{c}_{\mathbf{K}+\mathbf{k}'\downarrow}\right] \tag{1.17}$$

The result here obtained will be used various times in next chapters. Note that different Wick contraction schemes will be referred to as:

$$c^{\dagger}_{\mathbf{K}+\mathbf{k}\sigma}c^{\dagger}_{\mathbf{K}-\mathbf{k}\sigma'}c_{\mathbf{K}-\mathbf{k}'\sigma'}c_{\mathbf{K}+\mathbf{k}'\sigma} \qquad \text{Cooper contraction} \tag{1.18}$$
$$c^{\dagger}_{\mathbf{K}+\mathbf{k}\sigma}c^{\dagger}_{\mathbf{K}-\mathbf{k}\sigma'}c_{\mathbf{K}-\mathbf{k}'\sigma'}c_{\mathbf{K}+\mathbf{k}'\sigma} \qquad \text{Fock contraction} \tag{1.19}$$
$$c^{\dagger}_{\mathbf{K}+\mathbf{k}\sigma}c^{\dagger}_{\mathbf{K}-\mathbf{k}\sigma'}c_{\mathbf{K}-\mathbf{k}'\sigma'}c_{\mathbf{K}+\mathbf{k}'\sigma} \qquad \text{Hartree contraction} \tag{1.20}$$

When contracting in the Fock channel, a minus sign must be included.

**Local repulsion.** A somewhat identical argument can be carried out for the local interaction,

$$\hat{H}_U = U\sum_{\mathbf{r}\in\mathcal{S}}\hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}\downarrow}$$

The only difference here is made by the fact that the interaction is repulsive on-site, thus the final structure factor in the Fourier transform disappears as well as the minus sign,

$$\hat{H}_U \overset{\mathcal{F}}{=} \frac{U}{L_xL_y}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\uparrow}\hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\downarrow}\hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow}\hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \tag{1.21}$$

When contracting these fermionic operators, identical considerations as for the non-local counterpart (1.15) hold. Note the similarity of this Fourier transform with the o.s. sector of the non-local attraction, given in Eq. (1.15). With this, we conclude the general introduction to the model and pass to the setup of the Hartree-Fock (HF) algorithm.

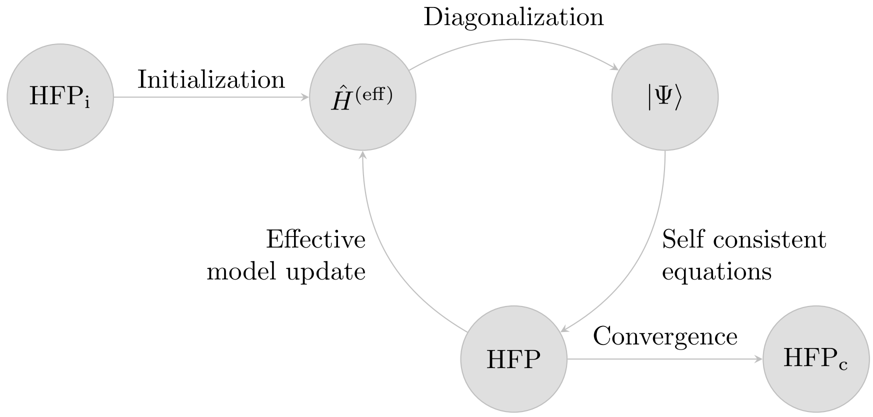



Figure 1.3 Schematization of the self-consistent Hartree Fock method described in Sec. 1.4. The effective model is solved, the solution is used to estimate self-consistently the Hartree Fock vector, which is then used to update iteratively the model itself. $\mathrm{HFP_i}$ indicates the initialzer HFPs vector, $\mathrm{HFP_c}$ the converged HFPs vector.

## 1.4 Iterative computational approach to MFT

This section is devoted to the explanation of the general iterative strategy we use to simulate the EHM through iterative Mean-Field Theory (iMFT). The simplicity of the algorithm is only possible thanks to the violent reduction of complexity MFT is able to reach. We use an iterative self-consistent scheme, thus solving a finite set of equation searching for a fixed point in parameters space for each phase.

### 1.4.1 Sketch of the Hartree-Fock iterative algorithm

The general sketch of a self-consistent Hartree-Fock algorithm in our context is simple enough. Start from a fermionic quartic hamiltonian,

$$\hat{H} = \sum_{\alpha\beta} A_{\alpha\beta}\hat{c}^\dagger_\alpha \hat{c}_\beta + \sum_{\alpha\beta\gamma\delta} B_{\alpha\beta\gamma\delta}\hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \hat{c}_\gamma \hat{c}_\delta + \text{h.c.} \tag{1.22}$$

where $A_{\alpha\beta}$, $B_{\alpha\beta\gamma\delta}$ are model parameters. We limit ourselves to quartic hamiltonians, but the idea can be extended. As discussed before, in this kind of context MFT is employed by applying Wick's theorem. This way, the quartic hamiltonian is effectively reduced to a quadratic hamiltonian,

$$\hat{H} \simeq \hat{H}^{(\mathrm{eff})} = \sum_{\alpha\beta} C_{\alpha\beta}\hat{c}^\dagger_\alpha \hat{c}_\beta + \sum_{\alpha\beta} D_{\alpha\beta}\hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta + \text{h.c.} \tag{1.23}$$

where now the new parameters, which we will refer to as Hartree-Fock parameters (HFP), depend on the model itself via the equations

$$\begin{aligned} C_{\alpha\beta} &= A_{\alpha\beta} + \sum_{\gamma\delta} \left[B_{\alpha\gamma\delta\beta} - B_{\alpha\gamma\beta\delta}\right] \langle \hat{c}^\dagger_\gamma \hat{c}_\delta \rangle \\ D_{\alpha\beta} &= \sum_{\gamma\delta} B_{\alpha\beta\gamma\delta} \langle \hat{c}_\gamma \hat{c}_\delta \rangle \end{aligned}$$

These equations are determined by many-body expectation values, which in general are computed by the means of thermal averages. A self consistent Hartree-Fock scheme goes as follows:

1. Apply MFT to the quartic hamiltonian, recovering its general theoretic effective form;
2. Impose selection rules on the parameters $C_{\alpha\beta}$, $D_{\alpha\beta}$, eventually eliminating those not respecting the symmetry sector we are working in. As an example: if we choose not to break charge conservation, set $\forall \alpha, \beta :\ D_{\alpha\beta} = 0$;

3. Choose a starting value and a coherent symmetry structure for the remaining parameters;
4. Enter the iterative scheme of Fig. 1.3:
   a) Based on the current HFPs, determine the value $\mu[C, D]$ giving the correct density;
   b) Diagonalize the effective hamiltonian $\hat{H}^{(\mathrm{eff})}[C, D] - \mu[C, D]\hat{N}$ and recover the ground-state $|\Psi[C, D]\rangle$;
   c) Compute the new HFPs $C'_{\alpha\beta}$, $D'_{\alpha\beta}$ via the self consistent equations;
   d) Initialize the new effective hamiltonian $\hat{H}^{(\mathrm{eff})}[C', D']$ and repeat the cycle until the HFPs have converged to stable values.

The stable values we are referring to in the last point are our solutions. The resulting vector is a fixed point in the parametric recursive map defined by the set of MFT self-consistent equations.

Of course, to work with any of these algorithms we need to know how to diagonalize the hamiltonian. In the context of the EHM at thermal equilibrium, being the phases we are simulating simple enough, we know analytically the diagonalized hamiltonian, and making use of this knowledge we are able to reduce the self consistent scheme to a simple iterative algorithm. An explicit example of HF algorithm used in present text, somewhat simplified due to the presence of a single HFP, is given in App. A.1.9.

### 1.4.2 Self-consistency and RMPs

As will be thoroughly discussed in the following derivations for each physical phase we treat, treating the EHM via MFT keeping under strict control the effects of Wick's theorem produces ground states whose parametric properties (band structures, gap functions, chemical potential) are controlled self-consistently by the model two-body interactions $\hat{H}_U$ and $\hat{H}_V$. Let us define the concept of Renormalized model parameter (RMP), indicating any of the model parameters that is effectively renormalized via interactions.

Let us propose a relevant example of RMP. Suppose that under MFT, for a given phase, the decomposition $\hat{H}_V$ has a net effect of renormalizing the hopping amplitude

$$t \overset{\hat{H}_V}{\to} \tilde{t}$$

In other words, the decomposition of $\hat{H}_V$ produces a free Fermionic hamiltonian whose diffusive parameter is $\tilde{t}$. We define $\tilde{t}$ as an RMP, being the result of the original model parameter $t$ plus the effect of interactions.

### 1.4.3 Fixed density simulations and the role of $\mu$

In the context of the present text, all simulations are performed at fixed density, determining self-consistently the value of the chemical potential. This means that, for a given step $i$ and the relative HFPs vector $\mathbf{v}_i$, and for a fixed density $n \in [0, 1]$, $\mu_i$ (the chemical potential at current step) is determined as

$$\mu_i \quad : \quad \min_{\mu} \left\{ \frac{1}{2L_x L_y} \sum_{\mathbf{k}} \sum_{\sigma} \langle \hat{n}_{\mathbf{k}\sigma} \rangle \big|_{\mathbf{v}=\mathbf{v}_i} - n \right\} \tag{1.24}$$

being $\mathbf{v} = \mathbf{v}_i$ the HFPs vector at current step. Averaging is performed over the ground state determined parametrically by the current value of the HFPs.

Hence, if we are able to perform averages and know both $\mathbf{v}_i$ and $n$, at step $i$ we infer the correct chemical potential. Now, as we will discuss $\tilde{\mu}$ will turn out to be an RMP itself, meaning that theoretically

$$\mu \to \mu + F[\mathbf{v}] \equiv \tilde{\mu}[\mathbf{v}]$$

being $F$ some functional of the HFPs. In other words, a common effect of interactions under MFT will be precisely to shift the chemical potential. At step $i$, keeping the density fixed, the chemical potential we are determining numerically and using in our equations is then $\tilde{\mu}_i$. In simpler words, the designed algorithm "does not know" about chemical potential renormalization and thus just determines the number that solves Eq. (1.24). But we know that this number is made of two contributions: the "physical" $\mu_i$ and the effective shift produced by the model itself, $F[\mathbf{v}_i]$. So, when it comes to $\mu_i$, this consideration becomes important.

It is the case of the ground state energy estimation. Let us abstract from the EHM and MFT, and consider a generic hamiltonian $\hat{H}$. To find the ground state $|\Psi\rangle$ means precisely to solve

$$\frac{\delta}{\delta \langle\Psi|} \langle\Psi|\hat{H}|\Psi\rangle = 0$$

Now, working with a fixed particles number $N$ means we are performing a constrained minimization (which is, taking the functional derivative) on the composite operator

$$\hat{H} - \mu(\hat{N} - N) \equiv \hat{K} + \mu N$$

being $\mu$ the Lagrange multiplier, $\hat{N}$ the number operator and $\hat{K} \equiv \hat{H} - \mu\hat{N}$. Finding the constrained ground state parametrized by a vector $\mathbf{v}$ now means to solve

$$\mathbf{v} \implies |\Psi[\mathbf{v}]\rangle \quad : \quad \begin{cases} \dfrac{\delta}{\delta \langle\Psi|} \langle\Psi|\hat{K} + \mu N|\Psi\rangle \Big|_{|\Psi\rangle = |\Psi[\mathbf{v}]\rangle} = 0 \\ \langle\Psi[\mathbf{v}]|\hat{N}|\Psi[\mathbf{v}]\rangle = N \end{cases}$$

In the first line $\mu N$ is derived to zero. The constraint is imposed in the second line. This means that, when it comes to determining the parametric ground state, $\hat{K}$ is functionally minimized. In this context $\mu$ (the Lagrange multiplier) sums with other effective terms arisen from algebraic manipulations of $\hat{H}$ giving a grand total of $\tilde{\mu}$. The number operator $\hat{N}$ average is performed over the many-body ground state $|\Psi[\mathbf{v}]\rangle$ with the total effective chemical potential being $\tilde{\mu}$, not $\mu$. However, when it comes to computing energy,

$$\begin{aligned} \langle\hat{H}\rangle &= \langle\Psi[\mathbf{v}]|\hat{K}|\Psi[\mathbf{v}]\rangle + \mu N \\ &= \Omega[\mathbf{v}] + (\tilde{\mu}[\mathbf{v}] - F[\mathbf{v}])N \end{aligned}$$

being $\Omega$ the expectation value (EV) of $\hat{K}$. In the last passage we wrote everything in terms of functionals of the HFPs vector $\mathbf{v}$. Using this relation, when the algorithm has converged, we can estimate the physical energy.

### 1.4.4 The reject-mix-iterate convergence scheme

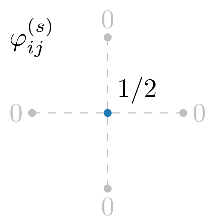
Up to now, we did not specify when we consider the algorithm to "have converged". From a purely theoretical point of view, considering the set of self-consistency equations provided by MFT as a nonlinear map in HFPs space, we consider the run finished when our algorithm, following a non-trivial path in HFPs space, comes close enough to a fixed point of the map; which is, a point which gets mapped onto itself. Explicitly, we consider the run over and converged whenever the following logic condition is satisfied for each component of $\mathbf{v}$:

$$\forall v_j \quad : \quad |v_j - \mathrm{A}_j(\mathbf{v})| \leq \Delta v_j$$

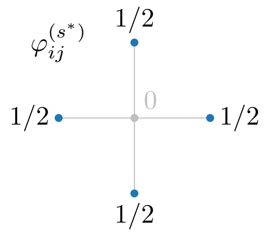
being $\mathrm{A}(\mathbf{v})$ the algorithmic image of the HFP vector $\mathbf{v}$, and $\mathrm{A}_j$ its $j$-th component. The specification of $\Delta v_j$ is clearly arbitrary. We are dealing in general with quantities of order $10^{-1}$, so convergence values of order $10^{-3} \div 10^{-4}$ will be generally sufficient.

When the acceptance condition is not satisfied, which is, when at least one component of $\mathbf{v}$ is mapped by the HF algorithm to a new image value for a total displacement larger than the respective component of $\Delta\mathbf{v}$, the step is rejected and iteration continues. This lets us define the mixing parameter $g \in [0, 1]$, which represents the deterministic fractional combination of the current step with its image value to obtain a new step,

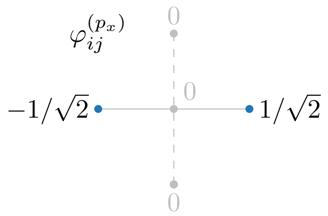
$$\mathbf{v}_{i+1} \equiv g\mathrm{A}\left(\mathbf{v}_i\right) + (1 - g)\mathbf{v}_i \tag{1.25}$$



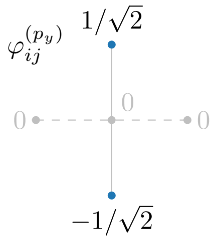

a | Local $s$-wave



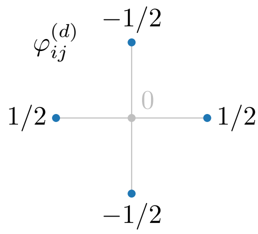

b | Extended $s$-wave.




c | $p_x$-wave.




d | $p_y$-wave.




e | $d_{x^2-y^2}$-wave.

Figure 1.4 | Form factors at different topologies, as listed in Tab. 1.5. In figures five sites are represented: the hub site $i$ and its four NN. Solid lines represent non-zero values for $\varphi_{\boldsymbol{\delta}}$, while dashed lines represent vanishing factors.

and becomes a simple but useful knob to tune algorithmic evolution. Its usefulness comes from the core concept that, even if physically there must exist a unique solution to the model (thus just one self-consistent HFP vector, eventually of all zeros) such a fixed point could be hard to reach following a pure iterative numeric scheme on a given map (which would be the $g = 1$ case of the above equation) for a variety of reasons. By moving "slower" (low $g$) with an accurate optimization of the mixing, as will be seen concretely in Sec. **??**, it is possible to reach convergence in a controlled way.

When it comes to HF iterative schemes of this kind, it can happen with some frequency that the algorithm remains stuck in an infinite loop between a set of points mathematically linked by the self-consistent map but not representing a physical solution (because of their reciprocal absolute distance in HFP space). In some other situations, evolutions could turn out to be rather slow. Because of this it is reasonable to define a parameter $p$, such that the number of iterations is kept $i \leq p$. If the algorithm failed to converge in $p$ steps, the point is saved as "`NaN`" (Not a number) and the algorithm moves to another simulation. Introducing this simple rule, runtime is kept under control and we gain an easy method to identify the regions of parameter space requiring more care – just read from the raw data where `NaN`s have been saved.

## 1.5 Square lattice spatial symmetry channels

In next sections we introduce spatial symmetry structures and discuss their relation with point groups. We will aways refer to various spatial structures as "symmetry channels", throughout the system is able to establish a given order parameter defined in space.

Looking at $\hat{H}$, the main difference between its two-body interactions $\hat{H}_U$ and $\hat{H}_V$ of course relies in locality (first) and non-locality (second). Both operators in principle are able to break $C_4$ symmetry, as will later be discussed extensively.

For the 2D square lattice, various spatial symmetries are possible. Let $i$ be a specific site, and consider a generic complex function $g_{ij}$ whose domain is given by the site itself and its four NNs

| Structure | Form factor | Graph |
|---|---|---|
| $s$-wave | $\varphi^{(s)}_{\mathbf{rr'}} = \delta_{\mathbf{rr'}}$ | Fig. 1.4a |
| Extended $s$-wave | $\varphi^{(s^*)}_{\mathbf{rr'}} = \frac{1}{2}\left[\delta_{\mathbf{r'=r+\underline{x}}} + \delta_{\mathbf{r'=r-\underline{x}}} + \delta_{\mathbf{r'=r+\underline{y}}} + \delta_{\mathbf{r'=r-\underline{y}}}\right]$ | Fig. 1.4b |
| $p_x$-wave | $\varphi^{(p_x)}_{\mathbf{rr'}} = \frac{1}{\sqrt{2}}\left[\delta_{\mathbf{r'=r+\underline{x}}} - \delta_{\mathbf{r'=r-\underline{x}}}\right]$ | Fig. 1.4c |
| $p_y$-wave | $\varphi^{(p_y)}_{\mathbf{rr'}} = \frac{1}{\sqrt{2}}\left[\delta_{\mathbf{r'=r+\underline{y}}} - \delta_{\mathbf{r'=r-\underline{y}}}\right]$ | Fig. 1.4d |
| $d_{x^2-y^2}$-wave | $\varphi^{(d)}_{\mathbf{rr'}} = \frac{1}{2}\left[\delta_{\mathbf{r'=r+\underline{x}}} + \delta_{\mathbf{r'=r-\underline{x}}} - \delta_{\mathbf{r'=r+\underline{y}}} - \delta_{\mathbf{r'=r-\underline{y}}}\right]$ | Fig. 1.4e |

Table 1.5 Form factors for a function defined on a square lattice, with a prefactor included for normalization. We use lattice vector notation here. The last column indicates the graph representation of each contribution given in Fig. 1.4. Subscript $x^2-y^2$ is omitted for notational clarity.

| Structure | Structure factor | Graph |
|---|---|---|
| $s$-wave | $\varphi^{(s)}_{\mathbf{k}} = 1$ | Fig. 1.4a |
| Extended $s$-wave | $\varphi^{(s^*)}_{\mathbf{k}} = \cos k_x + \cos k_y$ | Fig. 1.4b |
| $p_x$-wave | $\varphi^{(p_x)}_{\mathbf{k}} = i\sqrt{2}\sin k_x$ | Fig. 1.4c |
| $p_y$-wave | $\varphi^{(p_y)}_{\mathbf{k}} = i\sqrt{2}\sin k_y$ | Fig. 1.4d |
| $d_{x^2-y^2}$-wave | $\varphi^{(d)}_{\mathbf{k}} = \cos k_x - \cos k_y$ | Fig. 1.4e |

Table 1.6 Structure factors derived from the correlation structures of Tab. 1.5. The functions hereby defined are orthonormal, and are plotted in Fig- 1.5.

(the domain concides with the points of Fig. 1.2). The function can be written in the $C_4$ basis

$$g_{ij} = \sum_{\gamma} g^{(\gamma)} \varphi^{(\gamma)}_{ij}$$

where $g^{(\gamma)} \in \mathbb{C}$ are the $g_{ij}$ symmetry channels coefficients, while $\varphi^{(\gamma)}_{ij}$ are the form factors listed in Tab. 1.5 and plotted in Fig. 1.5, a simple orthonormal rearrangement of the harmonics basis.

When dealing with phases which respect translational symmetry, or at most reduce it to a weaker symmetry by shrinking the Brillouin Zone, it's useful to work in reciprocal space, as discussed in Sec. 1.3. As said, a particular phase can show specific spatial symmetries, and so will the Hartree-Fock parameters. The key point to notice is that self-consistency equations involve sums (or integrals) to be computed over fermionic states, which can be heavily simplified by employing said symmetries. Consider the form factors of Tab. 1.5, and their Fourier transforms of Tab. 1.6. They satisfy the relations:

$$\frac{1}{L_x L_y}\sum_{ij} \varphi^{(\gamma)}_{ij} \varphi^{(\gamma)*}_{ij} = \delta_{\gamma\gamma'} \qquad \frac{1}{L_x L_y}\sum_{\mathbf{k}} \varphi^{(\gamma)}_{\mathbf{k}} \varphi^{(\gamma)*}_{\mathbf{k}} = \delta_{\gamma\gamma'}$$

If these symmetries are developed by the HFPs, it's of no use to integrate over the entire BZ, since different regions linked by periodicity of the form factors will lead to redundant calculations.

Finally, a very useful detail is given by the following argument. Consider the equation

$$\cos(\delta k_x) + \cos(\delta k_y) = \cos k_x \cos k'_x + \sin k_x \sin k'_x + \cos k_y \cos k'_y + \sin k_y \sin k'_y$$

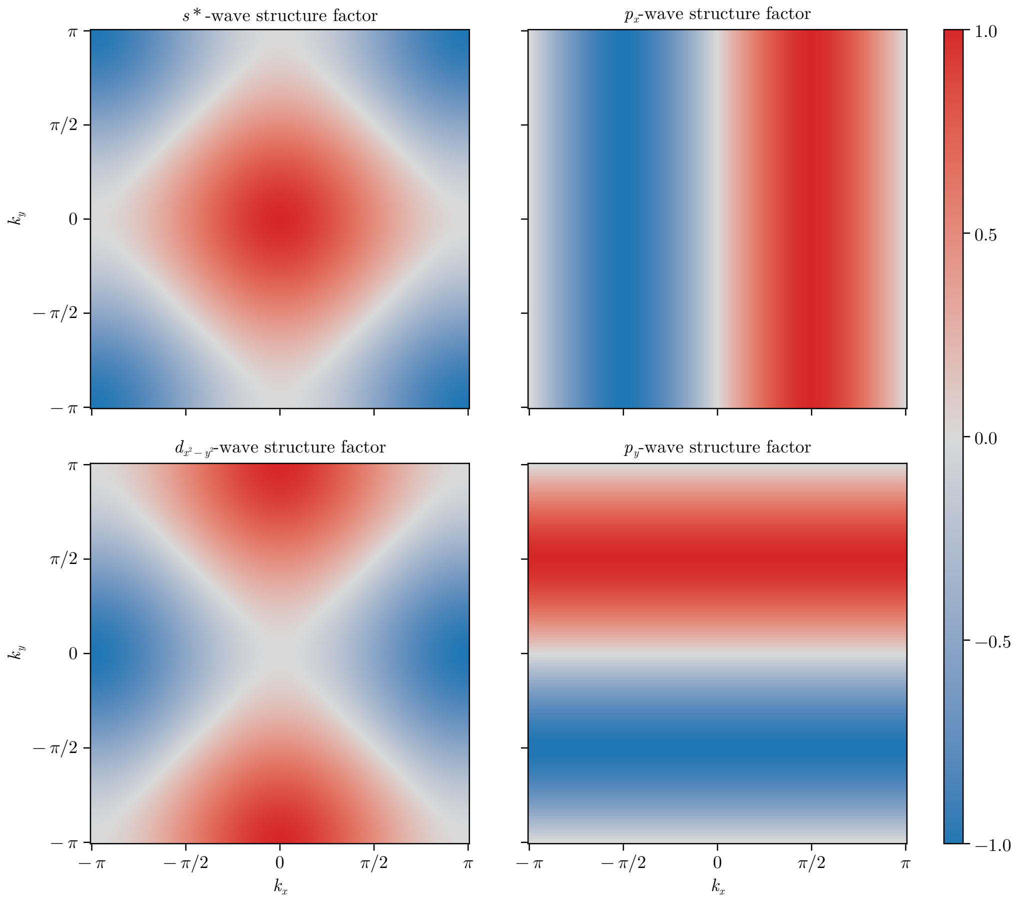



Figure 1.5 Representation of the various structure factors listed in Tab. 1.6 on the BZ. For the $s^*$ and $d_{x^2-y^2}$-wave factors, the real part is plotted; for the $p_\ell$ factors, the imaginary part is plotted.

For the sake of readability, the notations

$$c_\ell \equiv \cos k_\ell \qquad s_\ell \equiv \sin k_\ell \qquad c'_\ell \equiv \cos k'_\ell \qquad s'_\ell \equiv \sin k'_\ell$$

are used. Group the four terms above,

$$\underbrace{\left(c_x c'_x + c_y c'_y\right)}_{\text{Symmetric}} + \underbrace{\left(s_x s'_x + s_y s'_y\right)}_{\text{Anti-symmetric}} \tag{1.26}$$

The first two exhibit inversion symmetry for both arguments $\mathbf{k}$, $\mathbf{k}'$; the second two exhibit anti-symmetry. Decoupling the symmetric part, we obtain

$$c_x c'_x + c_y c'_y = \frac{1}{2}(c_x + c_y)(c'_x + c'_y) + \frac{1}{2}(c_x - c_y)(c'_x - c'_y)$$

which finally gives:

$$\begin{aligned}
\cos(\delta k_x) + \cos(\delta k_y) = {} & \frac{1}{2}(c_x + c_y)(c'_x + c'_y) && (s^*\text{-wave}) \\
& + s_x s'_x && (p_x\text{-wave}) \\
& + s_y s'_y && (p_y\text{-wave}) \\
& + \frac{1}{2}(c_x - c_y)(c'_x - c'_y) && (d_{x^2-y^2}\text{-wave})
\end{aligned}$$

or, in a more compact form

$$\cos(\delta k_x) + \cos(\delta k_y) = \frac{1}{2} \sum_{\gamma} \varphi_{\mathbf{k}}^{(\gamma)} \varphi_{\mathbf{k}'}^{(\gamma)*} \quad \text{for } \gamma \in \{s^*, p_x, p_y, d_{x^2-y^2}\} \tag{1.27}$$

$$= \frac{1}{2} \sum_{\gamma} \varphi_{\mathbf{k}}^{(\gamma)*} \varphi_{\mathbf{k}'}^{(\gamma)} \tag{1.28}$$

This decomposition will become particularly handy in later calculations.

# Appendix A

# Mean-Field Theory in Hubbard lattices

In this Appendix the MFT solutions to the conventional Hubbard hamiltonian of Eq. (??),

$$\hat{H} = -t \sum_{\langle ij \rangle} \sum_{\sigma} \left[ \hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} + \text{h.c.} \right] + U \sum_{i} \hat{n}_{i\uparrow} \hat{n}_{i\downarrow} \qquad t, U > 0 \tag{A.1}$$

are described. The discussion is limited to the two-dimensional square lattice. The results of ths appendix are relevant and used throughout the text as the groundwork to be extended by the means of MFT applied to the non-local part of the EHM.

## A.1 Antiferromagnetic phase

We work in a commensurate AF phase ordering, which is, we explicitly break symmetry by realizing a SDW phase with spin density unevenly distributed on sites with periodicity of two sites, each direction. Recalling the discussion of Sec. 1.1, AF long-range ordering we deal with here is described by the composite SSB

$$\mathrm{SU}(2) \overset{\mathrm{SSB}}{\to} \mathrm{U}^{z}(1)$$
$$\mathrm{T}^{(x)}_{1} \otimes \mathrm{T}^{(y)}_{1} \overset{\mathrm{SSB}}{\to} \mathrm{T}^{(x)}_{2} \otimes \mathrm{T}^{(y)}_{2}$$

Recalling Tab. 1.3, for such a physical phase the only Wick contraction channel available is the Hartree one. Let us proceed in such sense. As in Eqns. (??), (??), it is useful to define the density and AF fluctuation operators:

$$\hat{n}_{\mathbf{k}\sigma} \equiv \hat{c}^{\dagger}_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma} \tag{A.2}$$
$$\hat{\psi}_{\mathbf{k}\sigma} \equiv \hat{c}^{\dagger}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \hat{c}_{\mathbf{k}\sigma} \tag{A.3}$$

whose EVs are relevant order parameters for the normal and AF phases.

### A.1.1 MFT treatment of the Hartree channel

In terms of the physical magnetization $m$, the MFT Ansatz is given by

$$\langle \hat{n}_{\mathbf{r}\sigma} \rangle = n - (-1)^{x+y+\delta_{\sigma=\uparrow}} m \tag{A.4}$$

Decoupling the HM hamiltonian and substituting,

$$\begin{aligned} \hat{H} &\simeq -t \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \left[ \hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} + \text{h.c.} \right] + U \sum_{\mathbf{r}} \left[ \hat{n}_{\mathbf{r}\uparrow} \langle \hat{n}_{\mathbf{r}\downarrow} \rangle + \langle \hat{n}_{\mathbf{r}\uparrow} \rangle \hat{n}_{\mathbf{r}\downarrow} - \langle \hat{n}_{\mathbf{r}\uparrow} \rangle \langle \hat{n}_{\mathbf{r}\downarrow} \rangle \right] \\ &= -t \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} \left[ \hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} + \text{h.c.} \right] + nU \sum_{\mathbf{r}} \left[ \hat{n}_{\mathbf{r}\uparrow} + \hat{n}_{\mathbf{r}\downarrow} \right] - mU \sum_{\mathbf{r}} (-1)^{x+y} \left[ \hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow} \right] - E^{(\mathrm{AF})}_{\mathrm{H}/U} \end{aligned} \tag{A.5}$$

with $E^{(\mathrm{AF})}_{\mathrm{H}/U}$ the HM Hartree contraction energy,

$$\begin{aligned}
E^{(\mathrm{AF})}_{\mathrm{H}/U} &\equiv U\sum_{\mathbf{r}}\langle\hat{n}_{\mathbf{r}\uparrow}\rangle\langle\hat{n}_{\mathbf{r}\downarrow}\rangle \\
&= U\sum_{\mathbf{r}}\left(n+(-1)^{x+y}m\right)\left(n-(-1)^{x+y}m\right) \\
&= L_xL_y\times U\left(n^2-m^2\right)
\end{aligned} \tag{A.6}$$

By a Fourier transform, the phase factor of Eq. (A.5) can be absorbed in the destruction operator inside of $\hat{n}_{\mathbf{r}\sigma}$:

$$\begin{aligned}
\sum_{\mathbf{r}}(-1)^{x+y}\hat{n}_{\mathbf{r}\sigma} &= \sum_{\mathbf{r}}(-1)^{x+y}\hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}\sigma} \\
&\overset{\mathcal{F}}{=} \sum_{\mathbf{r}} e^{i\boldsymbol{\pi}\cdot\mathbf{r}}\frac{1}{\sqrt{L_xL_y}}\sum_{\mathbf{k}\in\mathrm{BZ}} e^{i\mathbf{k}\cdot\mathbf{r}}\hat{c}^\dagger_{\mathbf{k}\sigma}\frac{1}{\sqrt{L_xL_y}}\sum_{\mathbf{k}'\in\mathrm{BZ}} e^{-i\mathbf{k}'\cdot\mathbf{r}}\hat{c}_{\mathbf{k}'\sigma} \\
&= \sum_{\mathbf{k}\in\mathrm{BZ}}\sum_{\mathbf{k}'\in\mathrm{BZ}}\hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}'\sigma}\frac{1}{L_xL_y}\sum_{\mathbf{r}} e^{-i[\mathbf{k}'-(\mathbf{k}+\boldsymbol{\pi})]\cdot\mathbf{r}} \\
&= \sum_{\mathbf{k}\in\mathrm{BZ}}\hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma}
\end{aligned} \tag{A.7}$$

where $\boldsymbol{\pi}=(\pi,\pi)$. It follows:

$$mU\sum_{\mathbf{r}}(-1)^{x+y}\left[\hat{n}_{\mathbf{r}\uparrow}-\hat{n}_{\mathbf{r}\downarrow}\right] = mU\sum_{\mathbf{k}\in\mathrm{BZ}}\left[\hat{c}^\dagger_{\mathbf{k}\uparrow}\hat{c}_{\mathbf{k}+\boldsymbol{\pi}\uparrow}-\hat{c}^\dagger_{\mathbf{k}\downarrow}\hat{c}_{\mathbf{k}+\boldsymbol{\pi}\downarrow}\right]$$

Note that the Hartree channel renormalizes chemical potential,

$$nU\sum_{\mathbf{r}}\left[\hat{n}_{\mathbf{r}\uparrow}+\hat{n}_{\mathbf{r}\downarrow}\right] = nU\times\hat{N} \qquad\Longrightarrow\qquad \mu\to\tilde{\mu}\equiv\mu-nU$$

an effect that is present also in the EHM. The kinetic part of the hamiltonian transforms as in Eq. (1.13),

$$-t\sum_{\langle\mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}\left[\hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}'\sigma}+\mathrm{h.c.}\right] \overset{\mathcal{F}}{=} \sum_{\mathbf{k}\in\mathrm{BZ}}\sum_{\sigma}\epsilon_{\mathbf{k}}\hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma}$$

being $\epsilon_{\mathbf{k}}$ the bare bands of Eq. (1.14).

Take everything together. Using the property:

$$\begin{aligned}
\sum_{\mathbf{k}\in\mathrm{BZ}}\hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} &= \sum_{\mathbf{k}\in\mathrm{MBZ}}\left[\hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma}+\hat{c}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma}\hat{c}_{\mathbf{k}+2\boldsymbol{\pi}\sigma}\right] \\
&= \sum_{\mathbf{k}\in\mathrm{MBZ}}\left[\hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma}+\hat{c}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma}\hat{c}_{\mathbf{k}\sigma}\right]
\end{aligned} \tag{A.8}$$

and reduce Eq. (A.5) to its reciprocal form

$$\hat{H}\simeq\sum_{\mathbf{k}\in\mathrm{MBZ}}\sum_{\sigma}\epsilon_{\mathbf{k}}\left[\hat{n}_{\mathbf{k}\sigma}+\hat{n}_{\mathbf{k}+\boldsymbol{\pi}\sigma}\right]-mU\sum_{\mathbf{k}\in\mathrm{MBZ}}\sum_{\sigma}(-1)^{\delta_{\sigma=\downarrow}}\left[\hat{\psi}_{\mathbf{k}\sigma}+\hat{\psi}^\dagger_{\mathbf{k}\sigma}\right]+nU\hat{N}-E^{(\mathrm{AF})}_{\mathrm{H}/U}$$

### A.1.2 More on AF Ansatz

Before moving on, let us give a little detail on the AF Ansatz. The decoupling in Hartree sector of the density-density interaction can be treated with more care: denoting by d.c. the double counting terms we ignore,

$$\begin{aligned}
\hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}\downarrow}-\mathrm{d.c.} &= \hat{n}_{\mathbf{r}\uparrow}\langle\hat{n}_{\mathbf{r}\downarrow}\rangle+\langle\hat{n}_{\mathbf{r}\uparrow}\rangle\hat{n}_{\mathbf{r}\downarrow} \\
&= \hat{n}_{\mathbf{r}\uparrow}\frac{\langle\hat{n}_{\mathbf{r}\downarrow}\rangle+\langle\hat{n}_{\mathbf{r}\uparrow}\rangle}{2}+\hat{n}_{\mathbf{r}\uparrow}\frac{\langle\hat{n}_{\mathbf{r}\downarrow}\rangle-\langle\hat{n}_{\mathbf{r}\uparrow}\rangle}{2}+\frac{\langle\hat{n}_{\mathbf{r}\uparrow}\rangle+\langle\hat{n}_{\mathbf{r}\uparrow}\rangle}{2}\hat{n}_{\mathbf{r}\downarrow}+\frac{\langle\hat{n}_{\mathbf{r}\uparrow}\rangle-\langle\hat{n}_{\mathbf{r}\uparrow}\rangle}{2}\hat{n}_{\mathbf{r}\downarrow} \\
&= \frac{\langle\hat{n}_{\mathbf{r}\uparrow}\rangle+\langle\hat{n}_{\mathbf{r}\downarrow}\rangle}{2}\left(\hat{n}_{\mathbf{r}\uparrow}+\hat{n}_{\mathbf{r}\downarrow}\right)-\frac{\langle\hat{n}_{\mathbf{r}\uparrow}\rangle-\langle\hat{n}_{\mathbf{r}\downarrow}\rangle}{2}\left(\hat{n}_{\mathbf{r}\uparrow}-\hat{n}_{\mathbf{r}\downarrow}\right)
\end{aligned}$$

Our commensurate AF Ansatz essentially states that each site has the same spin-averaged density,

$$n \equiv \frac{\langle \hat{n}_{\mathbf{r}\uparrow} \rangle + \langle \hat{n}_{\mathbf{r}\downarrow} \rangle}{2} \qquad \text{is } \mathbf{r}\text{-independent}$$

while the spin averaged unbalance has a two-sites periodicity

$$m \equiv (-1)^{x+y} \frac{\langle \hat{n}_{\mathbf{r}\uparrow} \rangle - \langle \hat{n}_{\mathbf{r}\downarrow} \rangle}{2} \qquad \text{is } \mathbf{r}\text{-independent}$$

This way we reduce translational invariance. Of course, if the two quantities above are $\mathbf{r}$-independent indeed, it must hold

$$n = \frac{1}{L_x L_y} \sum_{\mathbf{r}} \frac{\langle \hat{n}_{\mathbf{r}\uparrow} \rangle + \langle \hat{n}_{\mathbf{r}\downarrow} \rangle}{2} \tag{A.9}$$

$$m = \frac{1}{L_x L_y} \sum_{\mathbf{r}} (-1)^{x+y} \frac{\langle \hat{n}_{\mathbf{r}\uparrow} \rangle - \langle \hat{n}_{\mathbf{r}\downarrow} \rangle}{2} \tag{A.10}$$

While from the first equation we gain an operational method to compute density, from the second we get a magnetization self-consistency equation.

### A.1.3 Nambu representation of the AF phase and diagonalization

We now map the reciprocal version of the HM onto a simpler problem. Throughout this section we use the notation $\epsilon_{\mathbf{k}} \to \epsilon_{\mathbf{k}\sigma}$ with identical meaning. Let the Nambu spinors:

$$\hat{\Psi}_{\mathbf{k}\sigma} \equiv \begin{bmatrix} \hat{c}_{\mathbf{k}\sigma} \\ \hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \end{bmatrix}$$

Let also the gap function:

$$\Delta_{\mathbf{k}\sigma} = (-1)^{\delta_{\sigma=\downarrow}} mU$$

which gives the matricial decomposition

$$\hat{H} \simeq \sum_{\mathbf{k}\in\mathrm{MBZ}} \sum_{\sigma} \hat{\Psi}^{\dagger}_{\mathbf{k}\sigma} h_{\mathbf{k}\sigma} \hat{\Psi}_{\mathbf{k}\sigma} - E^{(\mathrm{AF})}_{\mathrm{H}/U} \quad \text{being} \quad h_{\mathbf{k}\sigma} \equiv \begin{bmatrix} \epsilon_{\mathbf{k}\sigma} & -\Delta^{*}_{\mathbf{k}\sigma} \\ -\Delta_{\mathbf{k}\sigma} & -\epsilon_{\mathbf{k}\sigma} \end{bmatrix} \tag{A.11}$$

Notice: the Nambu hamiltonian is a $2 \times 2$ matrix over the MBZ – which is half the full BZ, coherently with a solution which essentially bipartites the lattice giving back a double sized unit cell.

The system ground-state is obtained by the means of a Bogoliubov rotation. The hamiltonian maps onto the simple one of a spin in a magnetic field (the gap is purely real here),

$$h_{\mathbf{k}\sigma} = \epsilon_{\mathbf{k}\sigma} \tau^{z} - \Delta_{\mathbf{k}\sigma} \tau^{x}$$

being $\tau^{\alpha}$ the Pauli matrices. Then, defining

$$\hat{s}^{\alpha}_{\mathbf{k}\sigma} \equiv \hat{\Psi}^{\dagger}_{\mathbf{k}\sigma} \tau^{\alpha} \hat{\Psi}_{\mathbf{k}\sigma} \quad \text{and} \quad \mathbf{b}_{\mathbf{k}\sigma} \equiv \begin{bmatrix} -\Delta_{\mathbf{k}\sigma} \\ 0 \\ \epsilon_{\mathbf{k}\sigma} \end{bmatrix}$$

one gets:

$$\hat{H} \simeq \sum_{\mathbf{k}\in\mathrm{MBZ}} \sum_{\sigma} \mathbf{b}_{\mathbf{k}\sigma} \cdot \hat{\mathbf{s}}_{\mathbf{k}\sigma} - E^{(\mathrm{AF})}_{\mathrm{H}/U} \quad \text{where} \quad \hat{\mathbf{s}}_{\mathbf{k}\sigma} = \begin{bmatrix} \hat{s}^{x}_{\mathbf{k}\sigma} \\ \hat{s}^{y}_{\mathbf{k}\sigma} \\ \hat{s}^{z}_{\mathbf{k}\sigma} \end{bmatrix} \tag{A.12}$$

The hamiltonian represents a system of spins subject to local magnetic fields, each tilted by an angle $\tan 2\theta_{\mathbf{k}\sigma} = \Delta_{\mathbf{k}\sigma}/\epsilon_{\mathbf{k}\sigma}$, as sketched in Fig. A.1. Evidently the following relations, also reported in Tab. **??**, hold:

$$\hat{s}^{x}_{\mathbf{k}\sigma} = \hat{\psi}_{\mathbf{k}\sigma} + \hat{\psi}^{\dagger}_{\mathbf{k}\sigma} \tag{A.13}$$

$$\hat{s}^{y}_{\mathbf{k}\sigma} = i\left(\hat{\psi}_{\mathbf{k}\sigma} - \hat{\psi}^{\dagger}_{\mathbf{k}\sigma}\right) \tag{A.14}$$

$$\hat{s}^{z}_{\mathbf{k}\sigma} = \hat{n}_{\mathbf{k}\sigma} - \hat{n}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \tag{A.15}$$

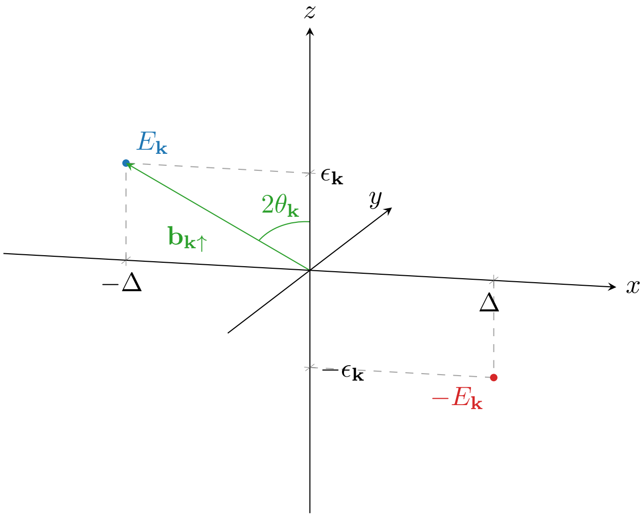



Figure A.1 | Pseudo-magnetic field originating from mean-field treatment of the square Hubbard hamiltonian. Here, only the $\sigma = \uparrow$ situation is plotted.

Let also:

$$\begin{aligned}\hat{\mathbb{I}}_{\mathbf{k}\sigma} &\equiv \hat{\Psi}^{\dagger}_{\mathbf{k}\sigma}\mathbb{I}_{2\times 2}\hat{\Psi}_{\mathbf{k}\sigma} \\ &= \hat{n}_{\mathbf{k}\sigma} + \hat{n}_{\mathbf{k}+\boldsymbol{\pi}\sigma}\end{aligned} \tag{A.16}$$

Diagonalization of each $h_{\mathbf{k}\sigma}$ is obtained trivially by a simple rotation around the $y$ axis:

$$d_{\mathbf{k}\sigma} = W_{\mathbf{k}\sigma} h_{\mathbf{k}\sigma} W^{\dagger}_{\mathbf{k}\sigma} \quad \text{with} \quad d_{\mathbf{k}\sigma} \equiv \begin{bmatrix} E_{\mathbf{k}\sigma} & \\ & -E_{\mathbf{k}\sigma}\end{bmatrix}$$

and where the diagonalization unitary matrix is simply a rotation by an angle $2\theta_{\mathbf{k}\sigma}$ around the $y$ axis, counter-clockwise

$$\begin{aligned}W_{\mathbf{k}\sigma} &\equiv e^{-i\theta_{\mathbf{k}\sigma}\tau^y} \\ &= \cos\theta_{\mathbf{k}\sigma} - i\sin\theta_{\mathbf{k}\sigma}\tau^y \\ &= \begin{bmatrix}\cos\theta_{\mathbf{k}\sigma} & -\sin\theta_{\mathbf{k}\sigma} \\ \sin\theta_{\mathbf{k}\sigma} & \cos\theta_{\mathbf{k}\sigma}\end{bmatrix}\end{aligned} \tag{A.17}$$

Eigenvalues are:

$$E_{\mathbf{k}\sigma} \equiv \sqrt{\epsilon^2_{\mathbf{k}\sigma} + |\Delta_{\mathbf{k}\sigma}|^2} \tag{A.18}$$

At any finite temperature, the ground-state of such a system is well-known: a thermal superposition of "down" spins with "up" spins. In the pseudo-spin picture, at each "site" ($\mathbf{k}\sigma$), the lowest-energy state is the one anti-aligned with the pseudo-magnetic field. It is useful now to define the thermal factors,

$$\mathrm{Th}^{(\pm)}_{\mathbf{k}\sigma}[\beta,\mu] \equiv f\left(-E_{\mathbf{k}\sigma};\beta,\mu\right) \pm f\left(E_{\mathbf{k}\sigma};\beta,\mu\right) \tag{A.19}$$

that will appear quite often. Then it's immediate to derive the various expectation values.

$$\langle \hat{s}^x_{\mathbf{k}\sigma}\rangle = \frac{\mathrm{Re}\{\Delta_{\mathbf{k}\sigma}\}}{E_{\mathbf{k}\sigma}}\mathrm{Th}^{(-)}_{\mathbf{k}\sigma}[\beta,\mu] \tag{A.20}$$

$$\langle \hat{s}^y_{\mathbf{k}\sigma}\rangle = \frac{\mathrm{Im}\{\Delta_{\mathbf{k}\sigma}\}}{E_{\mathbf{k}\sigma}}\mathrm{Th}^{(-)}_{\mathbf{k}\sigma}[\beta,\mu] \tag{A.21}$$

$$\langle \hat{s}^z_{\mathbf{k}\sigma}\rangle = -\frac{\epsilon_{\mathbf{k}\sigma}}{E_{\mathbf{k}\sigma}}\mathrm{Th}^{(-)}_{\mathbf{k}\sigma}[\beta,\mu] \tag{A.22}$$

Note that considering the gran canonical hamiltonian $\hat{K} \equiv \hat{H} - \mu\hat{N}$ and using Eq. (A.16) the diagonalizing matrix remains the same, being

$$-\mu\hat{N} = \sum_{\mathbf{k}\in\mathrm{MBZ}} \begin{bmatrix} -\mu & \\ & -\mu \end{bmatrix} \quad\Longrightarrow\quad W_{\mathbf{k}\sigma}\left[h_{\mathbf{k}\sigma} - \mu\mathbb{I}_{2\times 2}\right]W^{\dagger}_{\mathbf{k}\sigma} = d_{\mathbf{k}\sigma} - \mu\mathbb{I}_{2\times 2} \tag{A.23}$$

thus for non-null chemical potential (i.e. finite doping) the eigenvalues are just shifted, $\pm E_{\mathbf{k}} - \mu$. In other words, the new bands retain the same chemical potential of the bare bands. Notice also that the presence of an absolute value makes the eigenvalues independent of the $\sigma$ index. Eigenvector fermion operators are obtained simply as:

$$\hat{\Gamma}_{\mathbf{k}\sigma} = W_{\mathbf{k}\sigma}\hat{\Psi}_{\mathbf{k}\sigma} \tag{A.24}$$

The $\hat{\Gamma}$ spinor operators are effectively free fermionic spinor fields and as such must be treated. Explicitly,

$$\begin{aligned}\hat{\Gamma}_{\mathbf{k}\sigma} &= \begin{bmatrix} \cos\theta_{\mathbf{k}\sigma} & -\sin\theta_{\mathbf{k}\sigma} \\ \sin\theta_{\mathbf{k}\sigma} & \cos\theta_{\mathbf{k}\sigma} \end{bmatrix}\begin{bmatrix} \hat{c}_{\mathbf{k}\sigma} \\ \hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \end{bmatrix} \\ &= \begin{bmatrix} \cos\theta_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma} - \sin\theta_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \\ \sin\theta_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma} + \cos\theta_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \end{bmatrix} \equiv \begin{bmatrix} \hat{\gamma}^{(+)}_{\mathbf{k}\sigma} \\ \hat{\gamma}^{(-)}_{\mathbf{k}\sigma} \end{bmatrix}\end{aligned} \tag{A.25}$$

The consequence of Eq. (A.23) is interesting by itself: within this HF solution, the gap opens always at $E = 0$. In other words, the bands sketched in Fig. A.2b retain the same geometric features regardless of the doping, what changes is the bands width and the gap between them. The very key point here is that, being the two bands symmetric (thus containing each half of the electronic states) and defined over the MBZ, any finite doping $\delta > 0$ will produce a metallic behaviour, making chemical potential crossing the upper band. It is widely observed and confirmed by more refined numerical procedures, however, that AF establishes an insulating phase – here only reproduced at half-filling, where chemical potential lies within the gap. As will be detailed in A.1.8, the commensurate nature of AF SSB at wavevector $\boldsymbol{\pi}$ of this self-consistent solution is highly unstable and atypical at finite doping [17].

### A.1.4 Explicit zero-temperature AF ground state at half-filling

The AF ground state can be expressed easily at $n = 1/2$ and $\beta \to +\infty$. Let $|\Uparrow_{\mathbf{k}\sigma}\rangle$ and $|\Downarrow_{\mathbf{k}\sigma}\rangle$ be respectively the "up" and "down" $\mathbf{k}$-th pseudo-spin states. Let $\left|\overline{\Downarrow}_{\mathbf{k}\sigma}\right\rangle$ be the "down" state in the tilted frame. The ground state is given by

$$|\Psi\rangle \equiv \bigotimes_{\sigma}\bigotimes_{\mathbf{k}\in\mathrm{MBZ}}\left|\overline{\Downarrow}_{\mathbf{k}\sigma}\right\rangle \tag{A.26}$$

We can link the "down" states via an inverse rotation,

$$\begin{aligned}\left|\overline{\Downarrow}_{\mathbf{k}\sigma}\right\rangle &= e^{-i\theta_{\mathbf{k}\sigma}\hat{s}^{y}_{\mathbf{k}\sigma}}|\Downarrow_{\mathbf{k}\sigma}\rangle \\ &= \cos\theta_{\mathbf{k}\sigma}|\Downarrow_{\mathbf{k}\sigma}\rangle - i\sin\theta_{\mathbf{k}\sigma}\hat{s}^{y}_{\mathbf{k}\sigma}|\Downarrow_{\mathbf{k}\sigma}\rangle \\ &= \cos\theta_{\mathbf{k}\sigma}|\Downarrow_{\mathbf{k}\sigma}\rangle - \sin\theta_{\mathbf{k}\sigma}(\hat{s}^{+}_{\mathbf{k}\sigma} - \hat{s}^{-}_{\mathbf{k}\sigma})|\Downarrow_{\mathbf{k}\sigma}\rangle \\ &= \cos\theta_{\mathbf{k}\sigma}|\Downarrow_{\mathbf{k}\sigma}\rangle - \sin\theta_{\mathbf{k}\sigma}|\Uparrow_{\mathbf{k}\sigma}\rangle\end{aligned} \tag{A.27}$$

having we used

$$\hat{s}^{\pm}_{\mathbf{k}\sigma} \equiv \frac{\hat{s}^{x}_{\mathbf{k}\sigma} \pm i\hat{s}^{y}_{\mathbf{k}\sigma}}{2}$$

Use now Eqns. (A.16) and (A.15),

$$\frac{\hat{\mathbb{I}}_{\mathbf{k}\sigma} + \hat{s}^{z}_{\mathbf{k}\sigma}}{2} = \hat{n}_{\mathbf{k}\sigma} \qquad \frac{\hat{\mathbb{I}}_{\mathbf{k}\sigma} - \hat{s}^{z}_{\mathbf{k}\sigma}}{2} = \hat{n}_{\mathbf{k}+\boldsymbol{\pi}\sigma}$$

Evaluating the above operators on the "up" and "down" states, we get immediately

$$\frac{1}{2}\langle\Uparrow_{\mathbf{k}\sigma}|\hat{\mathbb{I}}_{\mathbf{k}\sigma}+\hat{s}^z_{\mathbf{k}\sigma}|\Uparrow_{\mathbf{k}\sigma}\rangle=1 \qquad \frac{1}{2}\langle\Uparrow_{\mathbf{k}\sigma}|\hat{\mathbb{I}}_{\mathbf{k}\sigma}-\hat{s}^z_{\mathbf{k}\sigma}|\Uparrow_{\mathbf{k}\sigma}\rangle=0$$
$$\frac{1}{2}\langle\Downarrow_{\mathbf{k}\sigma}|\hat{\mathbb{I}}_{\mathbf{k}\sigma}+\hat{s}^z_{\mathbf{k}\sigma}|\Downarrow_{\mathbf{k}\sigma}\rangle=0 \qquad \frac{1}{2}\langle\Downarrow_{\mathbf{k}\sigma}|\hat{\mathbb{I}}_{\mathbf{k}\sigma}-\hat{s}^z_{\mathbf{k}\sigma}|\Downarrow_{\mathbf{k}\sigma}\rangle=1$$

Thus, up to a phase

$$|\Uparrow_{\mathbf{k}\sigma}\rangle\sim\hat{c}^\dagger_{\mathbf{k}\sigma}|\Omega_{\mathbf{k}\sigma}\rangle \qquad |\Downarrow_{\mathbf{k}\sigma}\rangle\sim\hat{c}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma}|\Omega_{\mathbf{k}\sigma}\rangle$$

being $|\Omega_{\mathbf{k}\sigma}\rangle$ the $(\mathbf{k},\sigma)$ Hilbert subspace vacuum.

It follows that composing Eqns. (A.26) and (A.27) the ground state $|\Psi\rangle$ can be written as:

$$|\Psi\rangle=\bigotimes_{\sigma}\bigotimes_{\mathbf{k}\in\mathrm{MBZ}}\left[\sin\theta_{\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{k}\sigma}+e^{2i\zeta_{\mathbf{k}\sigma}}\cos\theta_{\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma}\right]|\Omega_{\mathbf{k}\sigma}\rangle \tag{A.28}$$

having we absorbed in the relative phase $e^{2i\zeta_{\mathbf{k}\sigma}}$ the two states phases differences as well as the original sign difference of Eq. (A.27). Now, inserting the explicit form of the ground state from equation above:

$$\begin{aligned}\langle\Downarrow_{\mathbf{k}\sigma}|\hat{\psi}_{\mathbf{k}\sigma}+\hat{\psi}^\dagger_{\mathbf{k}\sigma}|\Downarrow_{\mathbf{k}\sigma}\rangle&=\left(e^{2i\zeta_{\mathbf{k}\sigma}}+e^{-2i\zeta_{\mathbf{k}\sigma}}\right)\sin\theta_{\mathbf{k}\sigma}\cos\theta_{\mathbf{k}\sigma}\\&=\cos2\zeta_{\mathbf{k}\sigma}\sin2\theta_{\mathbf{k}\sigma}\end{aligned}$$

but from Eq. (A.13)

$$\begin{aligned}\langle\Downarrow_{\mathbf{k}\sigma}|\hat{\psi}_{\mathbf{k}\sigma}+\hat{\psi}^\dagger_{\mathbf{k}\sigma}|\Downarrow_{\mathbf{k}\sigma}\rangle&=\langle\Downarrow_{\mathbf{k}\sigma}|\hat{s}^x_{\mathbf{k}\sigma}|\Downarrow_{\mathbf{k}\sigma}\rangle\\&=\sin2\theta_{\mathbf{k}\sigma}\end{aligned}$$

since the pseudofield has no $y$ component. This implies necessarily

$$\cos2\zeta_{\mathbf{k}\sigma}\qquad\Longrightarrow\qquad\zeta_{\mathbf{k}\sigma}=k\pi\quad\text{with}\quad k\in\mathbb{Z}$$

and lets us conclude from Eq. (A.28)

$$|\Psi\rangle=\bigotimes_{\sigma}\bigotimes_{\mathbf{k}\in\mathrm{MBZ}}\left[\sin\theta_{\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{k}\sigma}+\cos\theta_{\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma}\right]|\Omega_{\mathbf{k}\sigma}\rangle \tag{A.29}$$

For low-temperature physics, we can expect the ground-state density matrix to be highly dominated by this BCS-like zero-temperature ground-state, which highlights the essential features of the system.

Finally, using Eq. (A.25), we check that correctly the ground-state describes a perfect degenerate $\gamma^{(-)}$ quasiparticles Fermi gas,

$$|\Psi\rangle=\bigotimes_{\sigma}\bigotimes_{\mathbf{k}\in\mathrm{MBZ}}\gamma^{(-)\dagger}_{\mathbf{k}\sigma}|\Omega_{\mathbf{k}\sigma}\rangle \tag{A.30}$$

The AF phase can be thought of as a long-range coordination between fermionic states related by a $\boldsymbol{\pi}$ shift in reciprocal space, producing a new free gas where all interactions have been "condensed" in these AF pairs.

### A.1.5 Spin degeneracy removal

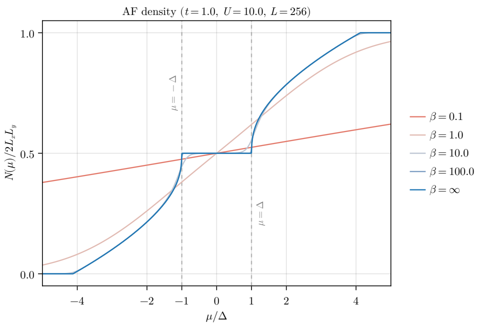
Up to now, we wrote explicitly the $\sigma$ index. However,

$$\epsilon_{\mathbf{k}\uparrow}=\epsilon_{\mathbf{k}\downarrow}\qquad\Delta_{\mathbf{k}\uparrow}=-\Delta_{\mathbf{k}\downarrow}$$

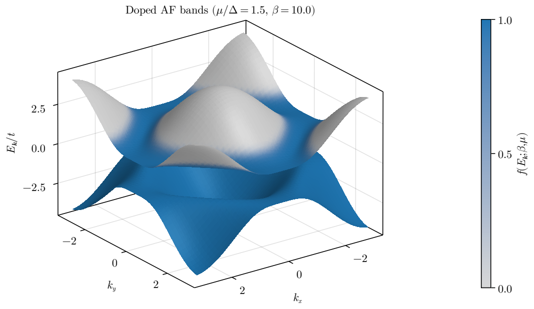
thus $E_{\mathbf{k}\uparrow}=E_{\mathbf{k}\downarrow}$. Since the gap function sign does has not a measurable impact, we might as well take the spin index as a pure degeneracy and from now on work in the $\sigma=\uparrow$ sector, dropping the $\sigma$ index. We can further simplify by dropping all indices from the gap function,

$$\Delta_{\mathbf{k}\uparrow}=mU\equiv\Delta$$

being the latter purely $s$-wave. This is a property of the pure HM: for the EHM, the gap acquires a non-trivial topological structure.




a | Electronic density.




b | Example doped bands.

Figure A.2 | Left panel: electronic density in the AF phase as a function of the chemical potential, measured in relation to the gap $\Delta = mU$, at various temperatures. Right panel: example doped bands, with $\mu/\Delta = 1.5$ set arbitrarily and temperature set to $\beta = 10$ to enhance smearing effects around FS. Single-state filling is proportional to color magnitude.

### A.1.6 Chemical potential and system density

Using the fact that

$$\left\langle \hat{\Gamma}^\dagger_{\mathbf{k}\sigma} \hat{\Gamma}_{\mathbf{k}\sigma} \right\rangle = \left\langle \hat{\Psi}^\dagger_{\mathbf{k}\sigma} \hat{\Psi}_{\mathbf{k}\sigma} \right\rangle$$

due to the $W_{\mathbf{k}\sigma}$ unitarity properties, the total number of electrons is given by (also recalling Eq. (A.9))

$$\begin{aligned} N &= \sum_{\mathbf{k}\sigma} \langle \hat{n}_{\mathbf{k}\sigma} \rangle \\ &= \sum_{\sigma} \sum_{\mathbf{k}\in\mathrm{MBZ}} \langle \hat{n}_{\mathbf{k}\sigma} + \hat{n}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \rangle \\ &= \sum_{\sigma} \sum_{\mathbf{k}\in\mathrm{MBZ}} \left\langle \hat{\Gamma}^\dagger_{\mathbf{k}\sigma} \hat{\Gamma}_{\mathbf{k}\sigma} \right\rangle \\ &= 2 \sum_{\mathbf{k}\in\mathrm{MBZ}} \mathrm{Th}^{(+)}_{\mathbf{k}}[\beta, \mu] \end{aligned} \tag{A.31}$$

being $f$ be the Fermi-Dirac distribution at inverse temperature $\beta$ and chemical potential $\mu$,

$$f(\epsilon; \beta, \mu) = \frac{1}{e^{\beta(\epsilon-\mu)} + 1}$$

The 2 prefactor appeared because of spin degeneracy. Since the gapped bands refer to the smaller MBZ, thus exhibit the periodicity

$$E_{\mathbf{k}} = E_{\mathbf{k}+\boldsymbol{\pi}}$$

it holds equivalently

$$N = \sum_{\mathbf{k}\in\mathrm{BZ}} \mathrm{Th}^{(+)}_{\mathbf{k}}[\beta, \mu] \tag{A.32}$$

as is obvious. The 2 prefactor was absorbed in doubling the integration region, MBZ $\to$ BZ. Next sections are devoted to simplify the above theoretical results in order to obtain an algorithmic estimation for the magnetization $m$, which is just the AF instability order parameter.

In Fig. A.2a the density $n(\mu)$ is plotted as a function of $\mu$. As is evident, at nearly zero temperature ($\beta \to \infty$) the system is half filled for $-\Delta < \mu < \Delta$. In order to obtain a finite doping, it must be $|\mu| > \Delta$; which means, the chemical potential needs to cross the electronic bands. This HF solution leads invariably to a rather unusual metallic AF at any finite doping.

### A.1.7 Self-consistency gap equation and magnetization

A convergence algorithm can be designed to find the Hartree-Fock solution to the model. Ultimately, we aim to extract $m$ self-consistently. Combining Eqns. (A.7), (A.8) and the definition of Eq. (A.3), we get

$$\frac{1}{2L_xL_y}\sum_{\mathbf{r}}(-1)^{x+y}\hat{n}_{\mathbf{r}\sigma} = \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\left[\hat{\psi}_{\mathbf{k}\sigma} + \hat{\psi}^{\dagger}_{\mathbf{k}\sigma}\right]$$

thus employing Eq. (A.10)

$$\begin{aligned} m &= \frac{1}{2L_xL_y}\sum_{\mathbf{r}}(-1)^{x+y}\langle\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}\rangle \\ &= \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\left\langle\left(\hat{\psi}_{\mathbf{k}\uparrow} + \hat{\psi}^{\dagger}_{\mathbf{k}\uparrow}\right) - \left(\hat{\psi}_{\mathbf{k}\downarrow} + \hat{\psi}^{\dagger}_{\mathbf{k}\downarrow}\right)\right\rangle \end{aligned}$$

Using now Eq. (A.13) we get the self-consistency equation for the HFP $m$,

$$\begin{aligned} m &= \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\left[\langle\hat{s}^x_{\mathbf{k}\uparrow}\rangle - \langle\hat{s}^x_{\mathbf{k}\downarrow}\rangle\right] \\ &= \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\frac{\mathrm{Re}\{\Delta_{\mathbf{k}\uparrow}\} - \mathrm{Re}\{\Delta_{\mathbf{k}\downarrow}\}}{E_{\mathbf{k}}}\mathrm{Th}^{(-)}_{\mathbf{k}}[\beta,\mu] \\ &= \frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\frac{\Delta}{E_{\mathbf{k}}}\mathrm{Th}^{(-)}_{\mathbf{k}}[\beta,\mu] \end{aligned} \tag{A.33}$$

The magnetization $m$ is the physical order parameter of the phase transition from the normal phase to the antiferromagnetic one. Within such phase transition, in fact, it is the expectation value

$$\langle\hat{c}^{\dagger}_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma}\rangle$$

that goes from zero (normal phase, free Fermi gas) to a non-zero value (AF). As is evident from equations above, $m$ is just its (only) $s$-wave component.

Consider specifically the half-filled situation and multiply Eq. (A.33) on both sides by $U$. Since due to symmetry $\mu = 0$, we have the finite-temperature self-consistent equation

$$\Delta = \frac{U}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\frac{\Delta}{\sqrt{\epsilon^2_{\mathbf{k}} + \Delta^2}}\tanh\left(\frac{\beta}{2}\sqrt{\epsilon^2_{\mathbf{k}} + \Delta^2}\right)$$

where we used the relation

$$\mathrm{Th}^{(-)}_{\mathbf{k}}[\beta,0] = \tanh\left(\frac{\beta E_{\mathbf{k}}}{2}\right)$$

In thermodynamic limit $L \to \infty$

$$\frac{1}{U} \simeq \int_{\mathrm{MBZ}}\frac{\mathrm{d}\mathbf{k}}{(2\pi)^2}\frac{1}{\sqrt{\epsilon^2_{\mathbf{k}} + \Delta^2}}\tanh\left(\frac{\beta}{2}\sqrt{\epsilon^2_{\mathbf{k}} + \Delta^2}\right)$$

Substitute the momentum integral with an integral over energies, in the low temperature limit, $\beta \to \infty$ (thus approximate the hyperbolic tangent with 1)

$$\frac{1}{U} \simeq \int_{\mathrm{bands}}\frac{\mathrm{d}\epsilon\rho(\epsilon)}{\sqrt{\epsilon^2 + \Delta^2}}$$

[To be continued. . . ]

### A.1.8 Instability of the commensurate AF solution and atypical metallic bands

A key point to be discussed is the known instability of the AF HF solution of the EHM. As is thoroughly discussed in [17], the solution we presented above is only stable at half-filling, becoming hardly unstable at infinitesimal doping. This is due to the commensurate nature of the established AF phase: we impose SSB to take place at wavevector $\boldsymbol{\pi}$, shrinking the BZ to exactly half its size and giving back on the newly defined MBZ two bands $\pm E_{\mathbf{k}}$. This procedure actually works well for the half-filled case: magnetization essentially depends on the scattering rate between points on the FS connected by a $\boldsymbol{\pi}$ vector. The half filled case, taking advantage of the natural nesting property of the bare bands,

$$\forall \mathbf{k} \in \partial\text{MBZ} : \; \epsilon_{\mathbf{k}} = \epsilon_{\mathbf{k}+\boldsymbol{\pi}}$$

(being $\partial$MBZ the boundary of the MBZ) exhibits a magnetization at infinitesimal interaction. For a many body state resulting from continuous SSB, to establish a given phase means to show the presence of a stable gapless Goldstone mode associated to the broken symmetry. Such a Goldstone mode is essentially a sharp peak at zero frequency for some dressed response function. For the AF phase, it follows from the reduced $\text{U}^z(1)$ rotational symmetry and is related to the proper transverse pseudo-spin to pseudo-spin response function, $2 \times 2$ matrix equation,

$$\tilde{\chi}_{\hat{\mathbf{S}}^-\hat{\mathbf{S}}^+}(\mathbf{k},\omega) = \frac{\chi_{\hat{\mathbf{S}}^-\hat{\mathbf{S}}^+}(\mathbf{k},\omega)}{1 - U\chi_{\hat{\mathbf{S}}^-\hat{\mathbf{S}}^+}(\mathbf{k},\omega)} \tag{A.34}$$

being $\chi$ the free transverse response. The pseudo-spin hereby considered is the one of Eq. (A.12). As is shown in [17], for the commensurate AF SSB at wavevector $\boldsymbol{\pi}$ the condition for the presence of a stable gapless mode reduces to a $2 \times 2$ matrix equation,

$$\text{Stability condition:} \qquad \mathbb{I}_{2\times 2} - U\left[\chi_{\hat{\mathbf{S}}^-\hat{\mathbf{S}}^+}(\boldsymbol{\pi},0)\right] \geq 0 \tag{A.35}$$

with $\left[\chi_{\hat{\mathbf{S}}^-\hat{\mathbf{S}}^+}(\boldsymbol{\pi},\omega)\right]$ the response function matrix obtained by considering a matrix response function for a two-sites unit cell. The above equation is to be interpreted as a matricial equation [Explain better..]. Considering the eigenvalue of largest amplitude of the matrix $\lambda_\delta$ at filling $\delta$, the one most prone to instabilities, it is found that

$$\begin{aligned}
\lambda_0(\mathbf{k}\to\boldsymbol{\pi},0) &= \frac{1}{U} && \text{(half filling)}\\
\lambda_\delta(\mathbf{k}\to\boldsymbol{\pi},0) &= \frac{1}{U} + \frac{2\epsilon_{\text{F}}}{\Delta}N(\epsilon_{\text{F}}) && \text{(finite doping)}
\end{aligned}$$

The reason for the distinction above is the intrinsic metallic nature of the system for infinitesimal doping. At half-filling the system is an insulator, therefore no Fermi energy is defined, as well as no FS. Evidently this implies

$$\begin{aligned}
&1 - U\lambda_0(\mathbf{k}\to\boldsymbol{\pi},0) = 0 \quad \text{(half filling)} && \Longrightarrow \quad \text{stable Goldstone mode}\\
&1 - U\lambda_\delta(\mathbf{k}\to\boldsymbol{\pi},0) < 0 \quad \text{(finite doping)} && \Longrightarrow \quad \text{unstable Goldstone mode}
\end{aligned}$$

For any finite doping, the commensurate AF phase becomes unstable and is destroyed by transverse spin fluctuations. Nevertheless, for the sake of completeness and because we noticed too late, we will study the metallic AF also at finite doping.

### A.1.9 Hartree-Fock algorithm results in reciprocal space

The above sections offer a self-consistency equation for the only HFP $m$; $W$ is determined by $\Delta = mU$ indeed. In an iterative HF fashion, Eq. (A.33) reads:

$$m_{i+1} = \frac{U}{L_x L_y}\sum_{\mathbf{k}\in\text{MBZ}} \frac{m_i}{2E_{\mathbf{k}}[\Delta_i]}\left[f(-E_{\mathbf{k}}[m_i];\beta,\mu) - f(E_{\mathbf{k}}[m_i];\beta,\mu)\right] \tag{A.36}$$

Then a self-consistent algorithm to search for a self consistent estimate of the magnetization may be sketched, following the general idea of Sec. 1.4:

1. Algorithm setup: initialize a counter $i = 1$ and choose:
   - the local repulsion $U/t$ or equivalently the hopping amplitude $t/U$;
   - the coarse-graining of the BZ, which is, fix $L_x$ and $L_y$. Then

     $$k_x = 2\pi n_x / L_x \qquad k_y = 2\pi n_y / L_y$$

     where $-L_j/2 \le n_j \le L_j/2$, $n_j \in \mathbb{Z}$;
   - the density $n$ or equivalently the doping $\delta \equiv n - 0.5$;
   - the inverse temperature $\beta$;
   - the maximum number of iterations $p \in \mathbb{N}$;
   - the mixing parameter $g \in [0, 1]$;
   - the tolerance parameters $\delta_m, \delta_n \in \mathbb{R}$ (respectively for the HFP $m$ and density);
2. Select a random starting value $m_i > 0$ for $i = 1$;
3. Find the optimal chemical potential $\mu$ as follows:

   a) Compute $\pm E_{\mathbf{k}}[\Delta_i, U]$;

   b) Define the function of the chemical potential $\mu$,

   $$\delta n(\mu; \beta, m_i, U) \equiv \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathrm{MBZ}} \left[ f\left(-E_{\mathbf{k}}[U, m_i]; \beta, \mu\right) + f\left(E_{\mathbf{k}}[U, m_i]; \beta, \mu\right) \right] - n$$

   c) Find the root of $\delta n$,

   $$\delta n(\mu_i) = 0$$

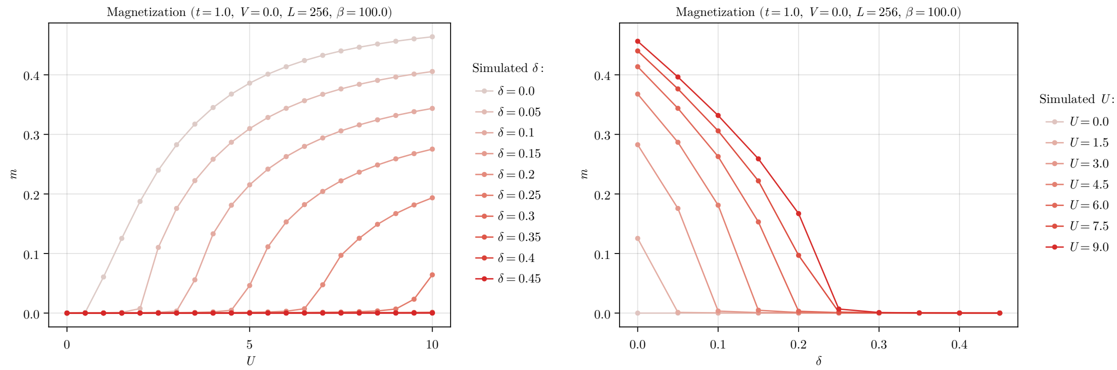
   Then $\mu_i$ is the chemical potential which, at step $i$ with HFP $m_i$, realizes the best approximation of density $n$;
4. Compute $m_{i+1}$ using Eq. (A.36) with chemical potential $\mu_i$ and update the counter, $i \to i+1$;
5. Check if $|m_{i+1} - m_i| \le \delta_m$:
   - If yes, halt;
   - If not: check if $i > p$
     - If yes, halt and consider choosing better tolerance and model parameters;
     - If not, define

       $$m_{i+1} : g m_{i+1} + (1 - g) m_i$$

       (logical assignment notation used) and repeat from step 2.

For the sake of completeness, we decided to run our algorithm also at finite doping, knowing from the previous discussion of Sec. A.1.8 that the results at $\delta > 0$ are unstable. The algorithm sketched hereby was ran with the following setup:

```
UU = [U for U in 0.5:0.5:20.0]        # Local repulsions
LL = [256]                            # Lattice sizes
dd = [d for d in 0.0:0.05:0.45]       # Dopings
bb = [100.0, Inf]                     # Inverse temperatures
p = 100                               # Max number of iterations
dm = 1e-4                             # Tolerance on magnetization
dn = 1e-2                             # Tolerance on density
g = 0.5                               # Mixing parameter
```




a | Local repulsion $U/t$ depencence.

b | Doping $\delta = n - 0.5$ dependence.

Figure A.3 | Plots of the zero-temperature AF instability order parameter $m$ as a function of both the local repulsion $U/t$ (Fig. A.3a) and the doping $\delta = n - 1/2$ (Fig. A.3b). [Insert new more refined data.]

In Fig. A.3 plots at $\beta = 100$ of the magnetization $m$ is reported. As is evident in Fig. A.3a, doping disrupts AF ordering; the horizontal asymptote lowers as $\delta$ gets bigger, a mere consequence of the fact that the free space at each single-particle state in $k$-space shrinks as density increases; also, for $\delta > 0$, a finite magnetization $m$ exists only for $U \geq U_c[\delta]$ (a critical value parametrically dependent on doping). Fig. A.3b shows the dependence of magnetization on doping at various fixed local repulsions $U$: once again, to dope the material disrupt AF ordering. Identical analysis have been performed for different temperatures, leading to analogous results as long as $\beta \gtrsim 10$ and to $m \simeq 0$ for higher temperatures (predictably).

## A.2 (Impossibility of a stable) superconducting phase

Let us now move to the discussion of superconductivity in the HM. As for the AF phase, we use the results and analyses of this section as groundwork for the more sophisticated case of SC phase in the EHM. We anticipate here that BCS-like superconductivity cannot develop in the pure HM, due to the inherently repulsive nature of $\hat{H}_U$. BCS theory admits, for an half filled model, finite Cooper fluctuations for an infinitesimal attractive interaction. However, as will be discussed, by the means of MFT it is perfectly possible and coherent to derive a self-consistent superconducting solution to the HM. The point here is that such self-consistent solution sets to zero Cooper fluctuations.

As done for the AF phase in Sec. A.1 and following the introductory discussion of Sec. [Missing], let us start from the discussion of the phase symmetries. We aim to describe a uniform-density superconductor,

$$\langle \hat{n}_{i\sigma} \rangle = n$$

taking into account Cooper RWCs,

$$\langle \hat{c}^\dagger_{i\sigma} \hat{c}^\dagger_{i\bar{\sigma}} \rangle$$

which, having we broken $U^c(1)$ charge symmetry, we allow to fluctuate to non-zero values. For the pure HM, we know from Tab. 1.3 that both Hartree and Cooper pairing channels contribute to the RWCs set. We handle them separately.

### A.2.1 MFT treatment of the Hartree channel

For the Hartree channel, knowing the deal with a uniform-density phase, we can re-use the derivation of Sec. A.1.1 simply by setting $m = 0$. We immediately get from Eq. (A.5):

$$\begin{aligned} \hat{H}_U &\simeq U \sum_{\mathbf{r}} \left[ \hat{n}_{\mathbf{r}\uparrow} \langle \hat{n}_{\mathbf{r}\downarrow} \rangle + \langle \hat{n}_{\mathbf{r}\uparrow} \rangle \hat{n}_{\mathbf{r}\downarrow} - \langle \hat{n}_{\mathbf{r}\uparrow} \rangle \langle \hat{n}_{\mathbf{r}\downarrow} \rangle \right] + (\text{Cooper channel}) \\ &= nU \times \hat{N} - E^{(\mathrm{SC})}_{\mathrm{H}/U} + (\text{Cooper channel}) \end{aligned} \tag{A.37}$$

thus chemical potential renormalization $\tilde{\mu} \equiv \mu - nU$ is present also for the SC phase. The contraction energy comes immediately from Eq. (A.6),

$$E^{(\mathrm{SC})}_{\mathrm{H}/U} = L_x L_y \times n^2 U \tag{A.38}$$

We now move to Cooper channel.

### A.2.2 MFT treatment of the Cooper channel

Recall the result of Eq. (1.21) which gave the Fourier-transformed version of the local repulsion, and perform a Cooper class Wick's contraction on it,

$$\begin{aligned}\hat{H}_U \simeq (\text{Hartree channel}) + \frac{U}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \Big[ &\langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \rangle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \\ &+ \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \langle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \rangle - \langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \rangle \langle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \rangle \Big]\end{aligned} \tag{A.39}$$

We are not breaking translational invariance, thus only Cooper fluctuations with net zero total momentum are allowed. This means only $\mathbf{K} = \mathbf{0}$ contributes. Define as in Eq. (**??**) the pairing operator

$$\hat{\phi}_{\mathbf{k}} \equiv \hat{c}_{-\mathbf{k}\downarrow} \hat{c}_{\mathbf{k}\uparrow} \tag{A.40}$$

and define $w_{\mathbf{q}}$ as in Eq. (**??**),

$$w_{\mathbf{q}} \equiv \langle \phi^\dagger_{\mathbf{q}} \rangle$$

Its $s$-wave part is given by:

$$w^{(s)} \equiv \frac{1}{L_x L_y} \sum_{\mathbf{q}\in \mathrm{BZ}} w_{\mathbf{q}} \tag{A.41}$$

Let $\hat{n}_{\mathbf{k}\sigma}$ as in Eq. (A.2). Eq. (A.37) reduces to

$$\hat{H}_U \simeq (\text{Hartree channel}) + \frac{U}{L_x L_y} \sum_{\mathbf{q},\mathbf{q}'} \Big[ \langle \hat{\phi}^\dagger_{\mathbf{q}} \rangle \hat{\phi}_{\mathbf{q}'} + \hat{\phi}^\dagger_{\mathbf{q}} \langle \hat{\phi}_{\mathbf{q}'} \rangle \Big] - E^{(\mathrm{SC})}_{\mathrm{C}/U} \tag{A.42}$$

being $E^{(\mathrm{SC})}_{\mathrm{C}/U}$ the HM Cooper contraction energy,

$$\begin{aligned}E^{(\mathrm{SC})}_{\mathrm{C}/U} &\equiv \frac{U}{L_x L_y} \sum_{\mathbf{q}\mathbf{q}'} \langle \hat{\phi}^\dagger_{\mathbf{q}} \rangle \langle \hat{\phi}_{\mathbf{q}'} \rangle \\ &= L_x L_y \times U \big| w^{(s)} \big|^2\end{aligned} \tag{A.43}$$

Let now

$$\Delta^{(s)} \equiv -U w^{(s)}$$

We finally reduce $\hat{H}_U$ to

$$\hat{H}_U \simeq (\text{Hartree channel}) - \sum_{\mathbf{q}\in \mathrm{BZ}} \Big[ \Delta^{(s)} \hat{\phi}_{\mathbf{q}} + \Delta^{(s)*} \hat{\phi}^\dagger_{\mathbf{q}} \Big] - E^{(\mathrm{SC})}_{\mathrm{C}/U} \tag{A.44}$$

which can be easily mapped on a pseudo-spin model.

Composing Eqns. (A.37) and (A.44), we get the full MFT hamiltonian

$$\begin{aligned}\hat{H} \simeq \sum_{\mathbf{k}\in \mathrm{BZ}} \sum_{\sigma} \epsilon_{\mathbf{k}\sigma} \Big[ \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma} - \hat{c}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma} \hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \Big] \\ - \sum_{\mathbf{q}\in \mathrm{BZ}} \Big[ \Delta^{(s)} \hat{\phi}_{\mathbf{q}} + \Delta^{(s)*} \hat{\phi}^\dagger_{\mathbf{q}} \Big] + nU \times \hat{N} - \Big[ E^{(\mathrm{SC})}_{\mathrm{H}/U} + E^{(\mathrm{SC})}_{\mathrm{C}/U} \Big]\end{aligned} \tag{A.45}$$

### A.2.3 Nambu representation of the SC phase and diagonalization

We now proceed precisely as in Sec. A.1.3. Define the SC Nambu spinor

$$\hat{\Phi}_{\mathbf{k}} \equiv \begin{bmatrix} \hat{c}_{\mathbf{k}\uparrow} \\ \hat{c}^{\dagger}_{-\mathbf{k}\downarrow} \end{bmatrix}$$

Define the pseudo-spin operator:

$$\hat{s}^{\alpha}_{\mathbf{k}} \equiv \hat{\Phi}^{\dagger}_{\mathbf{k}} \tau^{\alpha} \hat{\Phi}_{\mathbf{k}}$$

and the shifted bare bands,

$$\xi_{\mathbf{k}} \equiv \epsilon_{\mathbf{k}} - \tilde{\mu}$$

Notice we are using the chemical potential RMP. We employ from the start the spin-independency of the bare bands, as well as their pure $s^{*}$-wave structure, since no Fock interaction in the present context chan change that. The following chain holds:

$$\begin{aligned} \hat{n}_{\mathbf{k}\uparrow} + \hat{n}_{-\mathbf{k}\downarrow} &= \hat{c}^{\dagger}_{\mathbf{k}\uparrow}\hat{c}_{\mathbf{k}\uparrow} + \hat{c}^{\dagger}_{-\mathbf{k}\downarrow}\hat{c}_{-\mathbf{k}\downarrow} \\ &= \hat{c}^{\dagger}_{\mathbf{k}\uparrow}\hat{c}_{\mathbf{k}\uparrow} - \hat{c}_{-\mathbf{k}\downarrow}\hat{c}^{\dagger}_{-\mathbf{k}\downarrow} + \hat{\mathbb{I}}_{\mathbf{k}} \\ &= \hat{\Phi}^{\dagger}_{\mathbf{k}}\tau^{z}\hat{\Phi}_{\mathbf{k}} + \hat{\mathbb{I}}_{\mathbf{k}} \\ &= \hat{s}^{z}_{\mathbf{k}} + \hat{\mathbb{I}}_{\mathbf{k}} \end{aligned} \tag{A.46}$$

Then, considering the kinetic part of the gran-canonical hamiltonian,

$$\begin{aligned} \hat{H}_t - \tilde{\mu}\hat{N} &= \sum_{\sigma}\sum_{\mathbf{k}\in\mathrm{BZ}} \left(\epsilon_{\mathbf{k}\sigma} - \tilde{\mu}\right)\hat{n}_{\mathbf{k}\sigma} \\ &= \sum_{\mathbf{k}\in\mathrm{BZ}} \left[\xi_{\mathbf{k}}\hat{n}_{\mathbf{k}\uparrow} + \xi_{-\mathbf{k}}\hat{n}_{-\mathbf{k}\downarrow}\right] \\ &= \sum_{\mathbf{k}\in\mathrm{BZ}} \xi_{\mathbf{k}}\hat{s}^{z}_{\mathbf{k}} + E^{(\mathrm{SC})}_{\mathrm{a}} \end{aligned}$$

being $E^{(\mathrm{SC})}_{\mathrm{a}}$ the "anticommutation energy" coming from the kinetic term,

$$E^{(\mathrm{SC})}_{\mathrm{a}} \equiv \sum_{\mathbf{k}\in\mathrm{BZ}} \xi_{\mathbf{k}} \tag{A.47}$$

The scalar term in the above equation is essential for estimating correctly the phase ground state free energy density. The full MFT gran-canonical hamiltonian in Nambu representation is obtained including Eq. (A.42),

$$\hat{H} - \mu\hat{N} \simeq \sum_{\mathbf{k}\in\mathrm{BZ}}\sum_{\sigma} \hat{\Phi}^{\dagger}_{\mathbf{k}} h_{\mathbf{k}} \hat{\Phi}_{\mathbf{k}} + E^{(\mathrm{SC})}_{\mathrm{a}} - \left[E^{(\mathrm{SC})}_{\mathrm{H}/U} + E^{(\mathrm{SC})}_{\mathrm{C}/U}\right] \tag{A.48}$$

(note the non-renormalized chemical potential on the left) being

$$h_{\mathbf{k}} \equiv \begin{bmatrix} \xi_{\mathbf{k}} & -\Delta^{*}_{\mathbf{k}} \\ -\Delta_{\mathbf{k}} & -\xi_{\mathbf{k}} \end{bmatrix}$$

while the off-diagonal gap is a constant,

$$\Delta_{\mathbf{k}} \equiv \Delta^{(s)}$$

This hamiltonian behaves as an ensemble of spins in local magnetic pseudofields

$$\mathbf{b}_{\mathbf{k}} \equiv \begin{bmatrix} -\mathrm{Re}\{\Delta_{\mathbf{k}}\} \\ -\mathrm{Im}\{\Delta_{\mathbf{k}}\} \\ \xi_{\mathbf{k}} \end{bmatrix} \qquad \tan 2\theta_{\mathbf{k}} \equiv \frac{\xi_{\mathbf{k}}}{\sqrt{\xi^{2}_{\mathbf{k}} + |\Delta_{\mathbf{k}}|^{2}}} \qquad \tan 2\zeta_{\mathbf{k}} \equiv \frac{\mathrm{Im}\{\Delta_{\mathbf{k}}\}}{\mathrm{Re}\{\Delta_{\mathbf{k}}\}} \tag{A.49}$$

precisely as was for the AF phase, with the essential difference that now the chemical potential RMP lies inside the $z$ component of the pseudofield itself. This is a consequence of the Nambu spinors definition. We have the spin hamiltonian

$$\hat{H} - \mu\hat{N} \simeq \sum_{\mathbf{k}\in\mathrm{BZ}} \mathbf{b}_\mathbf{k}\cdot\hat{\mathbf{s}}_\mathbf{k} + E_\mathrm{a}^{(\mathrm{SC})} - \left[E_{\mathrm{H}/U}^{(\mathrm{SC})} + E_{\mathrm{C}/U}^{(\mathrm{SC})}\right] \quad \text{where} \quad \hat{\mathbf{s}}_\mathbf{k} = \begin{bmatrix}\hat{s}_\mathbf{k}^x\\ \hat{s}_\mathbf{k}^y\\ \hat{s}_\mathbf{k}^z\end{bmatrix} \tag{A.50}$$

As was for the AF phase, to diagonalize the hamiltonian means essentially to align the spin $z$ axis with the direction of the pseudofield. Then, we diagonalize the hamiltonian via a set of rotations similarly to Eq. (A.17). The main difference is that in general the pseudofield can have a $y$ component, thus the diagonalizing matrix is made of a sequence of two rotations, both counter-clockwise, the first around the $z$ axis by an angle $2\zeta_\mathbf{k}$ to place the pseudofield in the $xz$ plane, the second around the new $y$ axis by an angle $2\theta_\mathbf{k}$ to align the $z$ axis with the pseudofield,

$$\begin{aligned}W_\mathbf{k} &\equiv e^{-i\theta_\mathbf{k}\tau^y}e^{-i\zeta_\mathbf{k}\tau^z}\\ &= \begin{bmatrix}\cos\theta_\mathbf{k} & -\sin\theta_\mathbf{k}\\ \sin\theta_\mathbf{k} & \cos\theta_\mathbf{k}\end{bmatrix}\begin{bmatrix}e^{-i\zeta_\mathbf{k}} & \\ & e^{i\zeta_\mathbf{k}}\end{bmatrix}\\ &= \begin{bmatrix}\cos\theta_\mathbf{k}e^{-i\zeta_\mathbf{k}} & -\sin\theta_\mathbf{k}e^{i\zeta_\mathbf{k}}\\ \sin\theta_\mathbf{k}e^{-i\zeta_\mathbf{k}} & \cos\theta_\mathbf{k}e^{i\zeta_\mathbf{k}}\end{bmatrix}\end{aligned} \tag{A.51}$$

obtaining the diagonal matrix

$$d_\mathbf{k} \equiv W_\mathbf{k}h_\mathbf{k}W_\mathbf{k}^\dagger \quad \text{where} \quad d_\mathbf{k} \equiv \begin{bmatrix}E_\mathbf{k} & \\ & -E_\mathbf{k}\end{bmatrix}$$

and the bands

$$E_\mathbf{k} \equiv \sqrt{\xi_\mathbf{k}^2 + |\Delta_\mathbf{k}|^2}$$

Finally, the Nambu spinor for Bogoliubov quasiparticles is given similarly to Eq. (A.24)

$$\hat{\Gamma}_\mathbf{k} = W_\mathbf{k}\hat{\Phi}_\mathbf{k} \tag{A.52}$$

Explicitly,

$$\begin{aligned}\hat{\Gamma}_\mathbf{k} &= \begin{bmatrix}\cos\theta_\mathbf{k}e^{-i\zeta_\mathbf{k}} & -\sin\theta_\mathbf{k}e^{i\zeta_\mathbf{k}}\\ \sin\theta_\mathbf{k}e^{-i\zeta_\mathbf{k}} & \cos\theta_\mathbf{k}e^{i\zeta_\mathbf{k}}\end{bmatrix}\begin{bmatrix}\hat{c}_{\mathbf{k}\uparrow}\\ \hat{c}_{-\mathbf{k}\downarrow}^\dagger\end{bmatrix}\\ &= \begin{bmatrix}\cos\theta_\mathbf{k}e^{-i\zeta_\mathbf{k}}\hat{c}_{\mathbf{k}\uparrow} - \sin\theta_\mathbf{k}e^{i\zeta_\mathbf{k}}\hat{c}_{-\mathbf{k}\downarrow}^\dagger\\ \sin\theta_\mathbf{k}e^{-i\zeta_\mathbf{k}}\hat{c}_{\mathbf{k}\uparrow} + \cos\theta_\mathbf{k}e^{i\zeta_\mathbf{k}}\hat{c}_{-\mathbf{k}\downarrow}^\dagger\end{bmatrix} \equiv \begin{bmatrix}\hat{\gamma}_\mathbf{k}^{(+)}\\ \hat{\gamma}_\mathbf{k}^{(-)}\end{bmatrix}\end{aligned} \tag{A.53}$$

These operators act as free fermion fields on the Bogoliubov bands $\pm E_\mathbf{k}$.

### A.2.4 Explicit zero-temperature SC ground state at half-filling

Reasoning precisely as in Sec. A.1.4, we can extract the half-filling zero temperature ground state, commonly referred to as the "BCS state". Eqns. (A.26) and (A.27) hold equivalently (ignoring the $\sigma$ index and extending momentum tensor product to the entire BZ),

$$|\Psi\rangle = \bigotimes_{\mathbf{k}\in\mathrm{BZ}}\left[\cos\theta_\mathbf{k}|\Downarrow_\mathbf{k}\rangle - \sin\theta_\mathbf{k}|\Uparrow_\mathbf{k}\rangle\right]$$

Using now Eq. (A.46), we have

$$\begin{aligned}\langle\Downarrow_\mathbf{k}|\hat{s}_\mathbf{k}^z + \hat{\mathbb{I}}_\mathbf{k}|\Downarrow_\mathbf{k}\rangle &= 0\\ \langle\Uparrow_\mathbf{k}|\hat{s}_\mathbf{k}^z + \hat{\mathbb{I}}_\mathbf{k}|\Uparrow_\mathbf{k}\rangle &= 2\end{aligned}$$

so up to a phase

$$|\Downarrow_\mathbf{k}\rangle \sim |\Omega_\mathbf{k}\rangle \qquad |\Uparrow_\mathbf{k}\rangle \sim \hat{\phi}_\mathbf{k}^\dagger|\Omega_\mathbf{k}\rangle$$

Hence, we have

$$|\Psi\rangle = \bigotimes_{\mathbf{k}\in\mathrm{BZ}}\left[\cos\theta_\mathbf{k} + e^{2i\zeta_\mathbf{k}}\sin\theta_\mathbf{k}\hat{\phi}_\mathbf{k}^\dagger\right]|\Omega_\mathbf{k}\rangle \tag{A.54}$$

The argument $2\zeta_\mathbf{k}$ of the phase factor is precisely the phase of the gap function, as can be shown easily.

### A.2.5 Superconducting fluctuations suppression in the conventional HM

Let us report the SC analogous of the AF Eqns. (A.13) and (A.14):

$$\hat{s}^x_{\mathbf{k}} = \hat{\phi}_{\mathbf{k}} + \hat{\phi}^\dagger_{\mathbf{k}} \tag{A.55}$$

$$\hat{s}^y_{\mathbf{k}} = i\left(\hat{\phi}_{\mathbf{k}} - \hat{\phi}^\dagger_{\mathbf{k}}\right) \tag{A.56}$$

while the analogous of Eq. (A.15) was given in Eq. (A.46). Let also the thermal factor

$$\mathrm{Th}^{(\pm)}_{\mathbf{k}}[\beta] \equiv f\left(-E_{\mathbf{k}};\beta,0\right) \pm f\left(E_{\mathbf{k}};\beta,0\right) \tag{A.57}$$

and the EVs:

$$\langle \hat{s}^x_{\mathbf{k}} \rangle = \frac{\mathrm{Re}\{\Delta_{\mathbf{k}}\}}{E_{\mathbf{k}}}\mathrm{Th}^{(-)}_{\mathbf{k}}[\beta] \tag{A.58}$$

$$\langle \hat{s}^y_{\mathbf{k}} \rangle = \frac{\mathrm{Im}\{\Delta_{\mathbf{k}}\}}{E_{\mathbf{k}}}\mathrm{Th}^{(-)}_{\mathbf{k}}[\beta] \tag{A.59}$$

$$\langle \hat{s}^z_{\mathbf{k}} \rangle = -\frac{\xi_{\mathbf{k}}}{E_{\mathbf{k}}}\mathrm{Th}^{(-)}_{\mathbf{k}}[\beta] \tag{A.60}$$

Up to now, the derivation was general. However, it is immediate to show that for the pure HM the SC phase can never establish. Recall Eq. (A.41): explicitly

$$\begin{aligned} w^{(s)} &= \frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \langle \hat{\phi}^\dagger_{\mathbf{k}} \rangle \\ &= \frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \langle \hat{\Phi}^\dagger_{\mathbf{k}} \tau^+ \hat{\Phi}_{\mathbf{k}} \rangle \\ &= \frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \frac{\langle \hat{s}^x_{\mathbf{k}} \rangle + i\langle \hat{s}^y_{\mathbf{k}} \rangle}{2} \\ &= \frac{1}{2L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \frac{\Delta_{\mathbf{k}}}{E_{\mathbf{k}}}\mathrm{Th}^{(-)}_{\mathbf{k}}[\beta] \end{aligned} \tag{A.61}$$

Now: $\Delta_{\mathbf{k}} = \Delta^{(s)} = -Uw^{(s)}$, thus if we define

$$U_{\mathrm{c}}[\beta] \equiv \left[\frac{1}{2L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \frac{\mathrm{Th}^{(-)}_{\mathbf{k}}[\beta]}{E_{\mathbf{k}}}\right]^{-1} \tag{A.62}$$

(the reason for this notation will be clear in Sec. [Missing]) we end with the equation:

$$w^{(s)} = -\frac{U}{U_{\mathrm{c}}[\beta]}w^{(s)}$$

Since evidently $U_{\mathrm{c}}[\beta] > 0$ by definition (the thermal factor is negative by construction), the above equation can be solved in $\mathbb{C}$ only by $w^{(s)} = 0$. This implication terminates the argument: superconductivity cannot establish self-consistently at this level in the HM.

# Appendix B

# Mixing optimization in HM model

This short appendix is meant to give a practical example on a topic we often referenced to in text: how optimization of $g$, the mixing parameter defined in Sec. 1.4.4 and given operatively in Eq. (1.25) as

$$\mathbf{v}_{i+1} \equiv g\mathrm{A}\left(\mathbf{v}_i\right) + (1-g)\mathbf{v}_i$$

can be relevant in order to ensure convergence of an iterative HF scheme as presented in Sec. 1.4.1.

## B.1 Map analysis for $s$-wave superconductivity in HM

As we thoroughly discussed in Sec. A.2, conventional repulsive HM cannot exhibit BCS $s$-wave superconductivity within a MFT general scheme. This is proven trivially at the end of Sec. A.2.5, where we end up withe the self-consistency equation

$$w^{(s)} = -\frac{U}{U_{\mathrm{c}}} w^{(s)} \tag{B.1}$$

only solved at $U > 0$ by $w^{(s)} = 0$. Now, our aim is the following: given any initializer $w_0^{(s)}$, our iterative HF algorithm should give

$$w_0^{(s)} \quad \overset{\mathrm{Alg}}{\rightarrow} \quad \cdots \quad \overset{\mathrm{Alg}}{\rightarrow} \quad \sim 0$$

The point is the following: is this behaviour possible at any model parametrization?

Consider the nonlinear map of Eq. (B.1) as "seen by the algorithm": at step $i$, we compute the left side by using the value of $w^{(s)}$ at step $i-1$ (also for computing $U_{\mathrm{c}}[\beta]$). Let us drop the superscript $(s)$ for the sake of readability. Seen this way, we are effectively moving along the numeric succession

$$w_{i+1} = -g\left(\frac{U}{U_{\mathrm{c}}[\beta]}\right)_i w_i + (1-g)w_i$$

the subscript indicating iteration. In previous equation we made use of Eq. (1.25) to generate the $i+1$-th step from step $i$. Let now:

$$U_{\mathrm{s}} \equiv \left(\frac{1}{g} - 1\right) U_{\mathrm{c}}$$

$$U_{\mathrm{h}} \equiv \left(\frac{2}{g} - 1\right) U_{\mathrm{c}}$$

the subscript indicating, for a reason clear soon, a "soft boundary" (first) and an "hard boundary" (second). Let us derive some trivial facts true whenever $w_i > 0$:

1. It always holds $w_{i+1} \leq w_i$. This is simple to see: if $w_{i+1} > w_i$,

$$-g\left(\frac{U}{U_{\mathrm{c}}[\beta]}\right)_i w_i + (1-g)w_i > w_i$$

$$-g\left(\frac{U}{U_{\mathrm{c}}[\beta]}\right)_i w_i - gw_i > 0$$

which is impossible because all involved terms are positive.

2. For $0 < U < U_\mathrm{s}$, we have $0 < w_{i+1} < w_i$. Indeed $w_{i+1}$ is greater than zero when:

$$
\begin{aligned}
-g\left(\frac{U}{U_\mathrm{c}}\right)_i w_i + (1-g)w_i > 0 \\
\frac{1-g}{g} > \frac{U}{U_\mathrm{c}} \\
U_\mathrm{s} > U
\end{aligned}
$$

3. For $U_\mathrm{s} < U < U_\mathrm{h}$, we have $-w_i < w_{i+1} < 0$. From point above, we know $U > U_\mathrm{s}$ implies $w_{i+1} < 0$ by simply inverting inequalities. Moreover,

$$
\begin{aligned}
-g\left(\frac{U}{U_\mathrm{c}}\right)_i w_i + (1-g)w_i > -w_i \\
\frac{2-g}{g} > \frac{U}{U_\mathrm{c}} \\
U_\mathrm{h} > U
\end{aligned}
$$

4. For $U > U_\mathrm{h}$, we have $w_{i+1} < -w_i$. This is trivially implied by inverting inequalities in last point.

If we know $U_\mathrm{c}$, for any given $g$ we can predict the algorithmic image of a given step.

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